

Dual-task network with attention-guided U-net and gradient-based adaptive weighting optimization for fetal ultrasound plane classification and biometric measurement (DTNAU-Net)

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ARTICLE INFO

Keywords:

Fetal ultrasound imaging
Dual-task learning
Attention mechanism
Multi-objective optimization
Biometric measurement
Segmentation and classification

ABSTRACT

Automated fetal biometric analysis is essential for early diagnosis of growth-related abnormalities. This study presents the Dual-Task Network with Attention-Guided U-Net (DTNAU-Net), a novel deep learning architecture integrating Residual Blocks, Dual Attention Mechanisms, Multi-Scale Feature Extraction, and Gradient-Based Adaptive Weighting Optimization for simultaneous fetal ultrasound segmentation and classification of fetal ultrasound planes. Ablation studies demonstrated that each component significantly contributed to the model, and ultimately achieved Dice Index of 91.12% and classification accuracy of 94.85%. Comparative analysis against existing state-of-the-art deep learning models on the HC18 and Zenodo datasets confirmed DTNAU-Net's superiority, with statistically significant improvements in segmentation accuracy (Dice Index of $91.12\% \pm 0.20$) and classification performance (F1 Score of $94.30\% \pm 0.12$). Additionally, dual-task can reduce inference time by 0.12 sec/24 frames when used classification task to remove undesired frames vs applying segmentation-task on all the frames, making real-time ultrasound analysis more feasible. Statistical validation using Paired *t*-test, Wilcoxon Signed-Rank Test, and Friedman Test reinforced these findings. Furthermore, Explainable AI techniques using Shapley Additive Explanations and Local Interpretable Model-Agnostic Explanations confirmed DTNAU-Net's interpretability by highlighting its focus on critical anatomical structures. These results demonstrate the model's potential for robust, efficient, and clinically reliable fetal biometric assessment, aiding in the early diagnosis of conditions like intrauterine growth restriction (IUGR) and macrosomia.

1. Introduction

Ultrasound imaging is an indispensable tool in prenatal care, offering real-time, non-invasive visualization of fetal development and maternal health. Accurate identification and measurement of fetal body parts such as the head, abdomen, brain, femur, and thorax are critical for assessing fetal growth, estimating gestational age, and diagnosing

developmental anomalies. These measurements provide essential information for detecting conditions like intrauterine growth restriction (IUGR) and microcephaly, which demand timely medical intervention (Fiorentino et al., 2023; Plotka et al., 2021). Early and precise evaluations enable healthcare providers to address potential complications effectively, thereby improving neonatal outcomes (Zeng et al., 2021).

Despite advances in ultrasound technology, achieving consistent and

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accurate segmentation and classification of fetal ultrasound plane remains a significant challenge. Variations in fetal positioning, maternal conditions, and imaging artifacts can obscure anatomical structures, complicating the analysis process. Traditional approaches, which often involve manual or semi-automated methods, are time-consuming and prone to human error, leading to variability in diagnostic outcomes (Ronneberger et al., 2015; Zhou et al., 2018). These limitations underscore the pressing need for automated solutions that deliver robust, reproducible results while streamlining clinical workflows (Jha et al., 2019; Degala et al., 2024).

Recent developments in deep learning have transformed medical image analysis, enabling high-accuracy automation for segmentation and classification tasks. Architectures like U-Net and its extensions, such as U-Net++ and ResUNet++, have demonstrated exceptional performance in medical image segmentation by leveraging encoder-decoder structures that preserve critical spatial information (Luo et al., 2021; Myronenko, 2019). However, these models predominantly address single-task learning, focusing on either segmentation or classification. In prenatal care, a dual-task approach that integrates both tasks within a unified framework is essential for achieving comprehensive, efficient, and accurate analyses (Ridzuan et al., 2022).

To address these challenges, we propose DTNAU-Net, a Dual-Task Network with Attention-Guided U-Net, augmented by a gradient-based adaptive weighting optimization framework. Unlike traditional multi-objective optimization techniques, which rely on static parameter adjustments or computationally intensive methods like particle swarm optimization, the proposed approach dynamically balances segmentation and classification tasks during training. By leveraging performance metrics to adaptively weight the respective losses, the framework ensures efficient learning and minimizes task-specific trade-offs. This novel strategy is tailored to the complexities of fetal ultrasound imaging, where both segmentation and classification accuracy are equally critical.

Although residual learning, attention mechanisms, and multi-scale feature fusion have been explored independently in prior medical image segmentation studies, DTNAU-Net distinguishes itself through their task-coupled integration within a dual-task learning framework. In particular, the proposed gradient-based adaptive weighting strategy dynamically balances plane classification and anatomical segmentation losses based on gradient magnitudes, preventing task dominance and enabling stable multi-task optimization. This design allows global plane-level contextual information to guide local anatomical boundary refinement, which is especially beneficial for noisy and heterogeneous fetal ultrasound images.

Key Contributions:

- We present a computationally efficient DTNAU-Net that unifies the tasks of classification and segmentation to reduce overall segmentation time by incorporating fetal ultrasound plane (view) classification in ultrasound imaging.
- The model incorporates an attention-guided U-Net structure that improves focus on significant anatomical regions, enhancing the overall accuracy of the results.
- Gradient-based adaptive weighting optimization is applied that dynamically adjusts the importance of segmentation and classification losses during training, balancing task performance without the overhead of static multi-objective optimization techniques.
- An adaptive feature fusion strategy has been integrated to blend high-level and low-level features from the encoder and decoder, resulting in improved boundary precision and structure retention.
- Comprehensive testing on real-world ultrasound datasets shows that DTNAU-Net outperforms existing models in both classification and segmentation tasks.

DTNAU-Net represents a significant advancement in fetal ultrasound imaging, offering a robust and scalable solution for dual-task learning.

The integration of gradient-based adaptive weighting optimization addresses the limitations of static methods, enabling the model to dynamically adapt to the demands of complex clinical data. This approach ensures accurate segmentation of subtle anatomical structures while maintaining high classification performance, thereby enhancing the reliability of biometric measurements.

Clinically, the deployment of DTNAU-Net has the potential to transform prenatal diagnostics by automating routine tasks, reducing variability, and improving workflow efficiency. Automated, standardized analyses mitigate the inconsistencies inherent in manual assessments, facilitating consistent monitoring of fetal growth and early detection of potential complications (Milletari et al., 2016). Furthermore, the proposed framework is particularly suited to resource-constrained settings, where access to expert practitioners may be limited. By bridging gaps in healthcare delivery, DTNAU-Net supports equitable and reliable care for mothers and their babies (Zhang et al., 2022; Tajbakhsh et al., 2016).

In this paper, we present DTNAU-Net and its architecture, focusing on how the gradient-based adaptive weighting framework effectively balances the dual tasks of classification and segmentation. We provide detailed experimental results that show the model's performance improvements over existing approaches.

2. Related work

Medical image analysis has witnessed significant advancements with the development of deep learning models tailored for segmentation and classification tasks. The U-Net architecture, introduced by Ronneberger et al. (Ronneberger et al., 2015), marked a major milestone in medical image segmentation with its encoder-decoder design and skip connections, enabling the retention of essential contextual information for precise segmentation. Extensions such as U-Net++ and ResUNet++ have further enhanced segmentation capabilities by introducing nested skip pathways and residual connections, facilitating deeper feature learning and improved segmentation outcomes (Jha et al., 2019; Zhou et al., 2018). However, these models predominantly focus on single-task segmentation and do not integrate classification, limiting their effectiveness in applications like fetal ultrasound imaging, where comprehensive anatomical analysis is critical.

Multi-task learning (MTL) has emerged as a promising approach for addressing multiple related tasks by sharing learned representations, leading to improved performance and generalization while optimizing computational resources (Qu et al., 2022). Zhao et al. (Zhao et al., 2023) proposed a framework that combined segmentation and classification tasks, demonstrating the potential for MTL to outperform single-task models. However, these approaches often rely on fixed or pre-defined loss weights, which can lead to suboptimal performance when tasks exhibit varying complexities or priorities. This limitation is particularly evident in fetal ultrasound imaging, where segmentation and classification tasks often vary in difficulty depending on anatomical structures and imaging conditions.

Attention mechanisms have further advanced deep learning models for medical image analysis by allowing the network to focus on the most relevant regions of an image. The Attention U-Net, proposed by Oktay et al. (Oktay et al., 2018), introduced attention gates into the U-Net architecture, enhancing segmentation accuracy by emphasizing critical features while suppressing irrelevant regions. More recent models, such as TransAttUnet (Chen et al., 2024) and U-Net Transformer (Wang et al., 2022) have incorporated self-attention and transformer-based modules, improving feature selection and global context understanding. However, these models primarily focus on segmentation tasks and do not address the need for dynamic optimization of dual-task learning for both segmentation and classification.

Several studies have attempted to integrate attention mechanisms with multi-task learning frameworks to address complex imaging challenges. For example, Chen et al. (Chen et al., 2019) proposed a dual-task

network with attention mechanisms for brain tumor analysis, demonstrating improved performance in both segmentation and classification. Although these efforts represent progress, they often rely on static optimization strategies, such as fixed loss weighting, which fail to adapt dynamically to changing task performance during training. Furthermore, comprehensive approaches specifically designed for fetal ultrasound imaging remain limited.

To address these challenges, recent research has explored adaptive optimization strategies for dynamically balancing task performance. Gradient-based optimization techniques have gained attention for their ability to adjust loss weights during training based on real-time task performance, enabling models to prioritize tasks dynamically. While adaptive weighting has shown promise in general multi-task learning settings, its application to dual-task learning in fetal ultrasound imaging, particularly in integrating segmentation and classification with attention-guided architectures, remains underexplored.

Building upon these advancements, we propose DTNAU-Net, a Dual-Task Network with Attention-Guided U-Net, incorporating a gradient-based adaptive weighting optimization mechanism. Unlike conventional methods that rely on static or computationally intensive optimization techniques, our approach dynamically balances segmentation and classification tasks during training by adjusting loss weights based on task-specific performance metrics. By integrating attention mechanisms and dynamic optimization, DTNAU-Net provides a unified and efficient solution for fetal ultrasound imaging, addressing key limitations of existing models.

3. Methods

In this section, we present the design and components of the DTNAU-Net framework for the classification and segmentation of fetal ultrasound images. The network combines attention mechanisms and multi-objective optimization to achieve robust and accurate outcomes.

3.1. Network architecture

The overall architecture is illustrated in Fig. 1, the proposed DTNAU-Net architecture is an advanced attention-guided U-Net model designed for simultaneous classification and segmentation tasks. Built upon an

encoder-decoder framework, the model incorporates dual attention mechanisms and gradient-based adaptive weighting optimization to effectively balance task-specific learning objectives. The encoder extracts hierarchical features through a series of convolutional blocks, residual learning pathways, and attention gates, enabling robust feature extraction at multiple scales. The decoder reconstructs high-resolution segmentation maps using transposed convolutions, skip connections, and attention mechanisms to preserve fine-grained anatomical details. In DTNAU-Net, plane classification is therefore not used to gate or control segmentation outputs; rather, it serves as auxiliary supervision that regularizes shared feature representations and improves robustness, while segmentation and biometric estimation are performed independently from the learned features.

3.1.1. Encoder

The encoder in DTNAU-Net is designed to extract multi-scale, hierarchical features from input ultrasound images, which are critical for robust segmentation and classification. This section details how the encoder processes the input data through various blocks to capture both fine and abstract features.

- **Convolutional Block:** The encoder begins with convolutional blocks that consist of two convolutional layers followed by ReLU activation. These layers are responsible for extracting local features such as edges and textures from the input ultrasound images. This step is essential for capturing intricate details that contribute to the accurate segmentation of fetal anatomical structures. The use of ReLU activation introduces non-linearity, allowing the network to learn complex representations required for medical image analysis.
- **Residual Learning:** Residual connections are integrated into the convolutional blocks, enabling the model to learn residual mappings rather than direct mappings. This design helps to preserve gradient flow during training, preventing issues like vanishing gradients that can occur in deeper networks. These residual pathways allow the encoder to efficiently learn both low- and high-level features, enhancing the model's ability to represent subtle anatomical details accurately.
- **Down sampling Block:** Each convolutional block is followed by a max-pooling layer to down sample the feature maps, reducing their

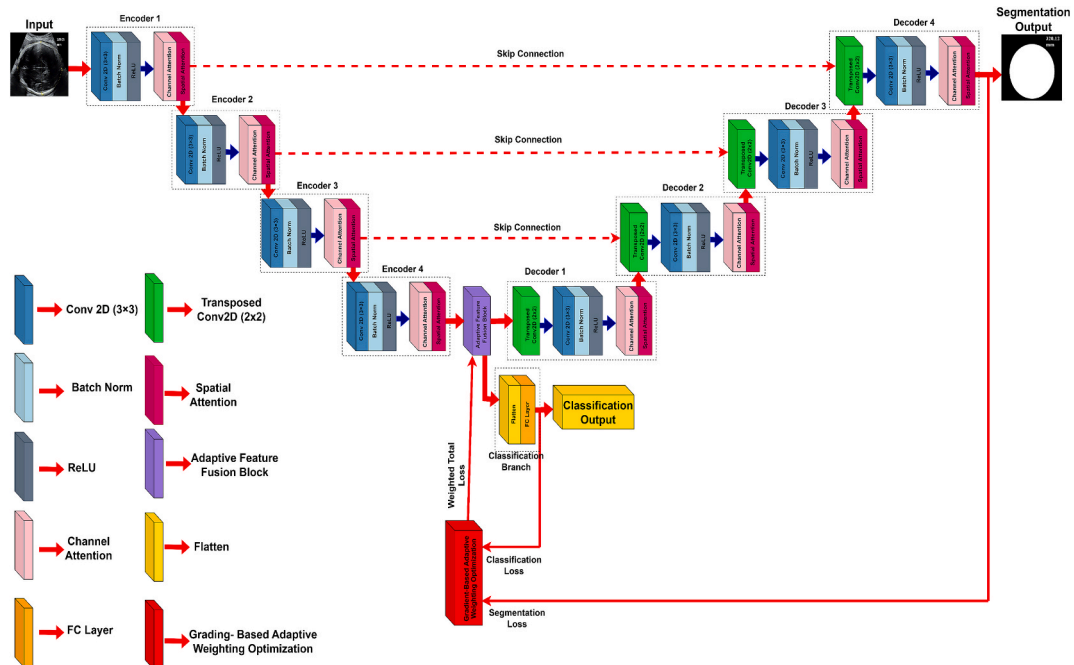


Fig. 1. Architecture of Dual-Task Network with Attention-Guided U-Net and Gradient-based adaptive weighting optimization.

spatial dimensions. This step condenses the information and allows the encoder to focus on higher-level contextual features while maintaining the essential details captured in earlier stages. The progressive down sampling helps the model balance local feature extraction and global context, which is crucial for distinguishing complex anatomical structures in ultrasound images.

- **Attention Gate Block:** Attention gates are strategically placed within the encoder to refine feature selection. These gates analyze the feature maps and prioritize the most relevant areas, enhancing the model's focus on essential anatomical regions while suppressing less important features. This mechanism ensures that the encoder emphasizes critical regions, such as the fetal head or limbs, which are vital for accurate segmentation and classification. By incorporating attention, the encoder improves its capacity to handle challenging image conditions, such as background noise and variable imaging quality.

The combination of these blocks ensures that the encoder in DTNAU-Net extracts comprehensive features at multiple scales, facilitating a balanced approach to image analysis that captures both fine details and broad contextual information.

3.1.2. Decoder

The decoder in DTNAU-Net plays a crucial role in reconstructing the segmentation map from the encoded feature representations. This section focuses on upsampling and restoring the spatial resolution to match the input image dimensions while preserving important anatomical features.

- **Up sampling Block:** The decoder employs transposed convolutional layers (deconvolutions) to increase the spatial resolution of the feature maps. These layers reverse the effect of down sampling, progressively restoring the image size to match the input. This process ensures that the final segmentation output maintains the original resolution, enabling detailed and accurate anatomical boundary reconstruction.
- **Skip Connections:** Skip connections are integrated between corresponding layers in the encoder and decoder to retain essential features and details. By passing features directly from the encoder to the decoder, the model preserves both low- and high-level information. This design helps in maintaining the fine details of anatomical structures, such as the borders of the fetal head or limbs, that might otherwise be lost during up sampling.
- **Residual Learning in Decoder:** The decoder also incorporates residual blocks to refine the up sampled feature maps. These blocks support continuous learning across layers, contributing to the precise segmentation of anatomical structures by maintaining feature continuity.
- **Attention Mechanisms in Decoder:** Attention gates are applied during decoding to focus on relevant features in the up sampled feature maps. This mechanism improves the accuracy of segmentation by ensuring the model emphasizes important regions while minimizing background noise and irrelevant features.

This multi-faceted approach allows the decoder in DTNAU-Net to reconstruct highly detailed segmentation maps while preserving the essential features learned in the encoder. The use of upsampling, skip connections, residual learning, and attention mechanisms creates a comprehensive framework for accurate and reliable image segmentation.

3.1.3. Adaptive feature fusion block

The Adaptive Feature Fusion Block in DTNAU-Net plays an essential role in merging multi-scale features from different encoder stages to create a unified and informative representation before passing them to the decoder. This block ensures that both low-level detailed features and

high-level abstract features are effectively combined, providing the decoder with a rich feature set for improved segmentation and classification outcomes.

- **Feature Aggregation:** The block aggregates feature maps extracted from various layers of the encoder. This process combines information at different resolutions, capturing fine details from shallow layers and broader contextual information from deeper layers. The aggregation can be performed through strategies such as weighted averaging or channel-wise concatenation, allowing the network to integrate diverse feature scales.
- **Transformation Layer:** Once the feature maps are aggregated, a transformation layer is applied to align their dimensions and prepare them for the decoder. This is achieved using a 1x1 convolutional layer that reduces the channel size while maintaining spatial information. The transformation step ensures that the combined feature map is consistent and compatible for subsequent processing.
- **Normalization and Activation:** The output of the transformation layer undergoes batch normalization followed by ReLU activation. Batch normalization stabilizes the training process by standardizing the output, reducing internal covariate shift. The ReLU activation introduces non-linearity, enabling the network to learn complex feature representations that contribute to accurate segmentation and classification.

The Adaptive Feature Fusion Block enhances the model's ability to utilize comprehensive information by blending detailed and abstract features effectively. This capability is crucial for handling the varying anatomical structures present in fetal ultrasound images, ensuring that the model has a complete understanding of the input image for precise segmentation and robust classification.

3.1.4. Gradient-Based adaptive weighting optimization Block

Unlike conventional multi-task networks that employ fixed or heuristically selected loss weights, DTNAU-Net adopts a gradient-based adaptive weighting mechanism to balance segmentation and plane classification objectives. At each training iteration, task-specific gradient magnitudes are monitored and used to dynamically adjust loss contributions, ensuring that neither task overwhelms the shared encoder. This strategy promotes balanced representation learning and improves convergence stability across datasets with heterogeneous anatomical appearances.

- **Gradient Monitoring:** The block monitors the gradients of the segmentation loss ($\mathcal{L}_{seg}$) and classification loss ($\mathcal{L}_{class}$) with respect to the shared network parameters. Gradient magnitudes provide a measure of task complexity, guiding the adjustment of task-specific weights.
- **Dynamic Weight Calculation:** The weights for segmentation and classification tasks are calculated based on their gradient magnitudes. Tasks with smaller gradients are assigned higher weights to prevent under-optimization, while tasks with larger gradients are down-weighted to avoid dominance. The weights are computed as:

$$w_{seg} = \frac{\exp(-\alpha g_{seg})}{\exp(-\alpha g_{seg}) + \exp(-\alpha g_{class})}$$

$$w_{class} = 1 - w_{seg}$$

where g_{seg} and g_{class} represent the gradient magnitudes, and α is a sensitivity parameter controlling the influence of gradient magnitudes on weight adjustment.

- **Composite Loss Integration:** The block combines segmentation and classification losses into a single composite loss:

$$\mathcal{L}_{total} = w_{seg} \cdot \mathcal{L}_{seg} + w_{class} \cdot \mathcal{L}_{class}$$

This formulation ensures balanced optimization of both tasks, adapting to their relative complexities during training.

- **Feedback Loop:** The weights are updated iteratively in each training step, creating a feedback loop that continuously refines the balance between tasks. This dynamic adjustment enhances the model's ability to adapt to varying task difficulties over time.

The Gradient-Based Adaptive Weighting Optimization Block improves DTNAU-Net's capacity to balance multi-task learning effectively, ensuring both segmentation and classification tasks achieve high performance. This capability is crucial in fetal ultrasound imaging, where accurate segmentation and classification are equally essential for reliable biometric measurements and clinical diagnostics.

To position the proposed strategy within the broader multi-task learning literature, we note that several adaptive loss balancing methods have been proposed, including uncertainty-based weighting, Gradient Normalization (GradNorm), Dynamic Weight Averaging (DWA), and gradient surgery approaches. Unlike uncertainty-based methods that introduce additional learnable variance parameters, and gradient normalization techniques that explicitly rescale gradients to enforce magnitude consistency across tasks, the proposed formulation computes task weights directly from instantaneous gradient magnitudes using an exponential normalization scheme. This design avoids additional trainable parameters, introduces negligible computational overhead, and maintains stable optimization in the presence of heterogeneous ultrasound image distributions. The method is therefore lightweight, numerically stable, and particularly suited for dual-task medical image learning where both objectives must remain balanced without introducing optimization instability.

3.1.5. Attention mechanisms block

The Attention Mechanisms Block in DTNAU-Net is designed to enhance the model's ability to focus on relevant features while suppressing less important information. Attention mechanisms help the network prioritize crucial areas in the image, improving both the segmentation and classification tasks. This block is integral to ensuring that the model pays attention to the most informative regions, such as specific anatomical structures in fetal ultrasound images.

- **Channel Attention:** The channel attention mechanism assigns different weights to feature maps based on their importance. This is achieved using global average pooling followed by fully connected layers to compute the weights, which are then applied to the feature maps. By emphasizing the most informative channels, the network can focus on feature maps that contribute significantly to identifying and segmenting relevant anatomical details.
- **Spatial Attention:** Spatial attention complements channel attention by generating an attention map over the spatial dimensions of the feature maps. This helps the model to focus on the most critical regions in the image. Spatial attention works by applying a convolutional operation followed by a sigmoid activation to generate an attention mask that highlights significant areas. This allows the model to concentrate on "where" the important features are located in the image.
- **Dual Attention Integration:** By combining channel and spatial attention mechanisms, DTNAU-Net achieves a dual attention structure that captures both "what" features to focus on (channel attention) and "where" they are located (spatial attention). This dual approach enhances the model's capacity to segment and classify with greater accuracy, especially in complex ultrasound images with varying noise levels and anatomical complexities.

The Attention Mechanisms Block plays a pivotal role in refining the

feature selection process, allowing DTNAU-Net to distinguish between relevant and irrelevant image regions effectively. This leads to improved performance in both segmentation and classification tasks by ensuring that the network attends to essential features that contribute to accurate medical analysis.

3.2. Dataset and data pre-processing

The proposed DTNAU-Net model was trained and evaluated using a combination of publicly available and custom datasets, as summarized in Table 1. The HC18 dataset, comprising 999 ultrasound images annotated for head circumference (HC), was used for training and testing HC segmentation. The Zenodo dataset, with 1,731 images annotated for HC, abdominal circumference (AC), biparietal diameter (BPD), and femur length (FL), provided a robust foundation for multi-parameter evaluation and validation. To enhance diversity and real-world applicability, a custom dataset of 3,000 ultrasound images was collected from Kamra Ultrasound Centre, Prayagraj, U.P., India, annotated for HC, AC, BPD, and FL, with strict adherence to patient privacy protocols.

Pre-processing steps were applied to ensure optimal model performance. All ultrasound images were resized to a standard dimension of 224×224 pixels to maintain uniformity. Pixel intensities were normalized by subtracting the mean and dividing by the standard deviation to stabilize training and improve convergence. Data augmentation techniques, including horizontal flipping, random rotation ($\pm 15^\circ$), and scaling ($0.9 - 1.1 \times$), were employed to increase dataset variability and reduce overfitting, simulating real-world imaging conditions. For segmentation tasks, binary masks were generated from the annotated regions to provide pixel-level supervision, while classification labels were assigned based on the corresponding fetal ultrasound imaging planes in each image. The datasets were evaluated using five-fold cross-validation performed at the patient level. In each fold, patients were partitioned into 60% training, 20% validation, and 20% testing subsets. The partitioning was strictly performed at the patient (study) level to ensure that no patient or associated frames appeared in more than one subset within a fold, thereby preventing data leakage. This process was repeated across five mutually exclusive folds, and the final reported results correspond to the mean $\pm$ standard deviation across folds.

3.3. Task formulation and label definition

The classification component of DTNAU-Net is formulated as a single-label multi-class fetal ultrasound plane (view) classification task. Each input image corresponds to one standardized biometric acquisition plane and is assigned a single ground-truth label.

The classification label set consists of three clinically standardized planes:

1. Head Plane (used for both Head Circumference (HC) and Biparietal Diameter (BPD) measurements)

Table 1

Summary of datasets used for training and evaluation of DTNAU-Net.

Dataset	Size	Annotations	Purpose
HC18 Dataset	999	Head Circumference (HC)	Training and testing for HC segmentation
Zenodo Dataset	1,731	HC, Abdominal Circumference (AC), Biparietal Diameter (BPD), Femur Length (FL)	Multi-parameter evaluation and validation
Custom Dataset (Kamra Ultrasound Centre, Prayagraj, India)	3,000	HC, AC, BPD, FL	Enhancing diversity and real-world applicability

2. Abdominal Plane (used for Abdominal Circumference (AC) measurement)
3. Femur Plane (used for Femur Length (FL) measurement)

Since each image in the evaluated datasets corresponds to a standardized biometric plane with a single annotated ground-truth label, the task is formulated as a single-label classification problem. Multi-label classification is not considered in this study.

In potential real-time deployment scenarios, “undesired frames” refer to blurred, transitional, or non-standard ultrasound views that do not correspond to these predefined biometric planes.

3.4. Implementation details

The DTNAU-Net model was implemented using Python and PyTorch, with training conducted on an NVIDIA T1000 GPU (8 GB VRAM) to ensure efficient computation. The Adam optimizer was employed with a learning rate of 1×10^{-6} , determined through experimentation across a range of values (1×10^{-4} to 1×10^{-7}). A batch size of 16 was chosen to balance memory constraints and training stability, while the model was trained for 100 epochs, with most performance gains observed within the first 80 epochs. For segmentation tasks, a combination of Dice loss and cross-entropy loss was used, whereas cross-entropy loss alone was applied for classification. Batch normalization was incorporated throughout the network to stabilize training and accelerate convergence.

To enhance training efficiency and generalization, a learning rate scheduler dynamically adjusted the learning rate based on validation performance, and early stopping was implemented to halt training if no improvement was observed over 15 consecutive epochs. Online data augmentation, including random rotation ($\pm 20^\circ$), horizontal flipping, and scaling (90–110%), was applied during training to increase dataset variability. The model’s performance was evaluated using a combination of segmentation and classification metrics, with the validation set used for hyperparameter tuning and the test set reserved for final performance assessment. This implementation strategy ensured robust training dynamics, minimized overfitting, and optimized DTNAU-Net for real-world clinical applications.

3.5. Fetal biometric measurements

The segmentation outputs from DTNAU-Net are post-processed to derive critical fetal biometric measurements, including HC, AC, BPD, and FL. The binary segmentation mask, initially at 224×224 pixels, is resized to match the original ultrasound image dimensions using bilinear interpolation. Binary thresholding (probability threshold = 0.6) is applied to remove false positives, followed by morphological operations (erosion and dilation with a 5×5 cross-shaped structuring element) to refine the segmented regions. A median blur filter (13×13 kernel) is then used to smooth edges, ensuring well-defined contours for accurate measurements.

For HC and AC, contours are detected, and an ellipse is fitted using a direct least square fitting method. The circumference is calculated using the Ramanujan approximation. BPD is measured as the short axis of the fitted ellipse, following an outer-to-outer approach, while FL is derived from the length of a bounding box fitted around the segmented femur region. Pixel-based measurements are converted to millimeters using the pixel size provided in the image metadata, ensuring clinically relevant outputs (Degala et al., 2024). These post-processing techniques and measurement methods enable DTNAU-Net to provide accurate and reliable biometric parameters, supporting healthcare professionals in assessing fetal growth and development.

3.6. Evaluation metrics

The performance of DTNAU-Net was evaluated using key metrics for

segmentation and classification tasks, critical for ensuring clinical reliability in fetal ultrasound analysis. For segmentation, the Dice Index (DI) (Zou et al., 2004) measured overlap accuracy between predicted and ground-truth masks, ensuring precise delineation of anatomical structures like the fetal head and abdomen. The Hausdorff Distance (HD) (Huttenlocher et al., 1993) quantified maximum boundary deviation to verify alignment with anatomical contours, while the Average Symmetric Surface Distance (ASSD) (Taha & Hanbury, 2015) assessed average boundary alignment. Mean Absolute Error (MAE) (Willmott & Matsuura, 2005) evaluated pixel-wise accuracy, and Percentage MAE (PMAE) (Zou et al., 2004) expressed this error relative to ground truth for clinical interpretability. The Boundary F1 Score (BF Score) (Csurka et al., 2013) balanced boundary pixel precision and recall, ensuring sharp anatomical delineation. For classification, Accuracy (Powers, 2020) measured overall correctness in fetal ultrasound plane (view) classification, Precision (Lennon et al., 2020) minimized false positives to avoid misdiagnoses, Recall (Sokolova & Lapalme, 2009) ensured detection of critical structures, and the F1 Score (Powers, 2020) balanced precision and recall for reliable clinical decisions.

These metrics collectively validate DTNAU-Net’s robustness in supporting fetal biometric measurements (e.g., HC, AC, BPD, FL) and clinical diagnostics. Detailed explanation and mathematical formulations for all metrics are provided in [Supplementary File](#).

3.7. Fairness and reproducibility of baseline comparisons

To ensure fairness and reproducibility of comparisons, all baseline models were retrained by the authors using publicly available official implementations or widely adopted open-source repositories. For nnU-Net and TransUNet, the official implementations provided by the original authors were used without architectural modifications.

For all models, identical preprocessing was enforced, including image resizing to 224×224 resolution, pixel intensity normalization, and patient-level five-fold cross-validation splits. The same data augmentation pipeline (random rotation $\pm 20^\circ$, horizontal flipping, scaling 90–110%) was applied uniformly across all models.

To eliminate performance differences arising from training protocols, all models were trained using the same optimizer (Adam), learning rate schedule, early stopping criterion, batch size (16), and number of training epochs (100). Only architecture-specific components intrinsic to each model were retained. Patch-based training was disabled for models such as nnUNet to ensure consistent input resolution across all methods.

Hyperparameter tuning was restricted to a limited and equal tuning budget for all models. Learning rates were selected from the same search range (1×10^{-4} to 1×10^{-7}), and the best validation performance within the cross-validation framework was reported.

No post-processing method was customized per model; the same biometric measurement pipeline described in [Section 3.5](#) was applied to all segmentation outputs.

These steps ensure that performance differences can be attributed to architectural design rather than preprocessing, training schedules, or optimization advantages.

3.8. Rationale for dual-task learning

The integration of plane classification and anatomical segmentation in DTNAU-Net is motivated by the complementary relationship between global anatomical understanding and local boundary delineation in fetal ultrasound imaging.

Although the fetal anatomical planes (HC, AC, FL, and BPD) exhibit visually distinct characteristics, ultrasound imaging is inherently affected by speckle noise, acoustic shadowing, low contrast, and probe-induced variability. Under such conditions, segmentation networks trained solely on pixel-level supervision may produce anatomically inconsistent or fragmented masks, particularly in ambiguous regions.

The auxiliary plane-classification branch addresses this limitation by enforcing global structural awareness during feature learning. Specifically:

- Head Circumference (HC) images emphasize circular skull morphology.
- Abdominal Circumference (AC) images exhibit elliptical abdominal contours.
- Femur Length (FL) images highlight elongated, high-echogenic linear bone structures.
- Biparietal Diameter (BPD) images exhibit symmetric cranial midline structures.

By supervising plane classification, the encoder is encouraged to capture morphology-aware representations that encode anatomical priors specific to each view. These global contextual features guide the segmentation decoder toward anatomically plausible predictions and reduce cross-structure misinterpretation.

Formally, the joint optimization objective is defined as:

$$L_{total} = \lambda_{seg}L_{seg} + \lambda_{cls}L_{cls}$$

where L

L_{seg} denotes the segmentation loss and L_{cls} denotes the plane classification loss. The weighting coefficients λ_{seg} and λ_{cls} balance the contribution of both tasks.

Beyond contextual guidance, the dual-task formulation acts as an implicit regularizer. Multi-task learning constrains the shared encoder to learn representations that are simultaneously discriminative at the global (plane-level) and local (pixel-level) scales. This reduces overfitting to speckle artifacts and improves generalization across heterogeneous datasets acquired from different scanners and clinical settings.

The effectiveness of this design choice is further validated through ablation experiments presented in Section 4.10.5, where removal of the classification branch leads to a measurable degradation in segmentation performance.

4. Results and discussion

All results presented in Sections 4.1–4.7 are derived from five-fold cross-validation conducted at the patient level. For each fold, training, validation, and testing partitions were created using a 60%–20%–20% split of patients. Performance metrics were computed independently on the test subset of each fold, and the final reported values represent the mean $\pm$ standard deviation across folds. Statistical analyses were performed using fold-wise performance values.

4.1. Comparison of classification with existing methods

The classification performance of DTNAU-Net was assessed against state-of-the-art models, using five-fold cross-validation across the HC18, Zenodo, and Kamra Ultrasound Centre datasets. The results of this evaluation are summarized in Table 2.

DTNAU-Net achieved the highest classification accuracy of $94.85 \pm 0.15\%$, significantly outperforming TransUNet ($93.90 \pm 0.18\%$) and FetSAM ($93.50 \pm 0.20\%$). In addition to accuracy, the model demonstrated exceptional precision and recall values, indicating its ability to minimize both false positives and false negatives effectively. The F1 Score of $94.30 \pm 0.12\%$ highlights the model's balanced performance across all classification metrics, ensuring reliable identification of fetal ultrasound planes across varying conditions and datasets.

The results underscore DTNAU-Net's robust classification framework, which integrates an attention-guided U-Net structure with multi-objective optimization. Models such as Attention U-Net and U-Net++ showed strong results but were limited in their ability to balance complex classification tasks with segmentation. On the other hand, advanced

Table 2

Comparison of Classification Performance Metrics across Baseline and State-of-the-Art Models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
U-Net	89.50 ± 0.40	89.20 ± 0.35	88.80 ± 0.30	89.00 ± 0.32
U-Net++	89.90 ± 0.38	89.60 ± 0.33	89.40 ± 0.28	89.50 ± 0.30
ResUNet++	90.30 ± 0.35	90.00 ± 0.30	89.70 ± 0.28	89.85 ± 0.29
Attention U-Net	91.10 ± 0.30	90.80 ± 0.25	90.50 ± 0.22	90.65 ± 0.23
Deeplabv3+	91.80 ± 0.28	91.50 ± 0.24	91.20 ± 0.20	91.35 ± 0.22
Double U-Net	92.30 ± 0.25	92.00 ± 0.20	91.80 ± 0.18	91.90 ± 0.19
MSRF-Net	93.00 ± 0.22	92.70 ± 0.18	92.50 ± 0.16	92.60 ± 0.17
FetSAM	93.50 ± 0.20	93.30 ± 0.18	93.10 ± 0.15	93.20 ± 0.16
FetalNet	93.80 ± 0.18	93.60 ± 0.16	93.50 ± 0.14	93.55 ± 0.15
TransUNet	93.90 ± 0.18	93.70 ± 0.15	93.60 ± 0.14	93.65 ± 0.15
nnUNet	93.70 ± 0.18	93.40 ± 0.15	93.30 ± 0.14	93.35 ± 0.15
Proposed DTNAU-Net	94.85 ± 0.15	94.50 ± 0.12	94.30 ± 0.10	94.30 ± 0.12

models like FetSAM and TransUNet struggled to maintain consistent performance across datasets, particularly in noisy or complex imaging scenarios.

Clinically, the high classification accuracy of DTNAU-Net ensures reliable identification of fetal ultrasound planes, which is critical for tasks like biometric measurement and anomaly detection. Accurate classification reduces the burden on sonographers and radiologists by automating routine image analysis, allowing them to focus on higher-level diagnostic tasks. This level of precision is especially beneficial in resource-constrained settings, where access to skilled professionals may be limited.

4.2. Comparison of segmentation with existing methods

The segmentation performance of DTNAU-Net was evaluated against state-of-the-art models, including U-Net (Ronneberger et al., 2015), U-Net++ (Zhou et al., 2018), ResUNet++ (Jha et al., 2019), Attention U-Net (Oktay et al., 2018), Deeplabv3+ (Chen et al., 2018), Double U-Net (Jha et al., 2020), MSRF-Net (Srivastava et al., 2022), FetSAM (Alzubaidi et al., 2024), FetalNet (Plotka et al., 2021), TransUNet (Chen et al., 2021), and nnUNet (Isensee et al., 2021), using five-fold cross-validation across the HC18, Zenodo, and Kamra Ultrasound Centre datasets. A detailed comparison of these models is presented in Table 3.

DTNAU-Net consistently demonstrated superior performance, achieving a Dice Index of $91.12 \pm 0.20\%$, surpassing TransUNet ($89.00 \pm 0.40\%$) and nnUNet ($88.80 \pm 0.50\%$). Its Hausdorff Distance of 5.03 ± 0.15 mm was the lowest among the compared models, indicating precise boundary delineation. The Boundary F1 Score of $88.50 \pm 0.20\%$ further underscores its effectiveness in identifying edge pixels accurately. Moreover, DTNAU-Net achieved the lowest MAE of 0.90 ± 0.01 mm and PMAE of $4.40 \pm 0.10\%$, highlighting its ability to produce segmentations closely aligned with ground truth measurements, essential for clinical applications.

The results highlight that DTNAU-Net addresses the challenges faced by other models in maintaining boundary integrity and capturing fine anatomical details. For example, U-Net and ResUNet++ exhibited higher variability in performance, struggling in cases with complex anatomical structures or noise. Advanced models like Attention U-Net and Deeplabv3+ demonstrated improvements but fell short in

Table 3

Comparison of Segmentation Performance Metrics across Baseline and State-of-the-Art Models.

Model	Dice Index (%)	HD (mm)	Boundary F1 (%)	MAE (mm)	PMAE (%)
U-Net	84.50 ± 0.85	7.80 ± 0.65	81.20 ± 0.75	1.42 ± 0.08	6.80 ± 0.50
U-Net++	85.20 ± 0.80	7.50 ± 0.60	82.30 ± 0.70	1.40 ± 0.07	6.50 ± 0.45
ResUNet++	85.60 ± 0.80	7.20 ± 0.50	82.90 ± 0.65	1.39 ± 0.06	6.40 ± 0.45
Attention U-Net	86.30 ± 0.75	6.90 ± 0.55	84.50 ± 0.60	1.32 ± 0.05	6.00 ± 0.40
Deeplabv3+	86.90 ± 0.65	6.70 ± 0.45	85.00 ± 0.55	1.28 ± 0.04	5.70 ± 0.30
Double U-Net	87.10 ± 0.50	6.50 ± 0.40	85.30 ± 0.45	1.22 ± 0.03	5.50 ± 0.25
MSRF-Net	88.20 ± 0.45	5.80 ± 0.35	86.70 ± 0.30	1.10 ± 0.02	5.00 ± 0.20
FetSAM	88.90 ± 0.40	5.60 ± 0.30	87.00 ± 0.35	1.05 ± 0.03	4.90 ± 0.20
FetalNet	89.20 ± 0.35	5.50 ± 0.25	87.10 ± 0.25	1.03 ± 0.02	4.80 ± 0.20
TransUNet	89.00 ± 0.40	5.60 ± 0.35	86.90 ± 0.30	1.04 ± 0.03	4.80 ± 0.25
nnUNet	88.80 ± 0.50	5.70 ± 0.30	86.80 ± 0.40	1.06 ± 0.03	4.90 ± 0.30
Proposed DTNAU-Net	91.12 ± 0.20	5.03 ± 0.15	88.50 ± 0.20	0.90 ± 0.01	4.40 ± 0.10

leveraging multi-scale features effectively. DTNAU-Net's architecture, integrating adaptive feature fusion and multi-objective optimization, proved instrumental in achieving robust segmentation performance

across all datasets.

Clinically, the precision of DTNAU-Net's segmentations translates to accurate biometric measurements, such as head circumference (HC), abdominal circumference (AC), and femur length (FL). These measurements are critical for diagnosing developmental anomalies like intrauterine growth restriction (IUGR) and microcephaly. The ability to reduce boundary errors and variability ensures consistent outputs across different operators and settings, enhancing clinical decision-making. Furthermore, its robust performance under varying levels of noise and complexity reinforces its applicability in diverse real-world scenarios, making DTNAU-Net a valuable tool for prenatal diagnostics.

4.2.1. Head circumference (HC) segmentation

Fig. 2 illustrates HC segmentation across models for a representative fetal ultrasound subject, with the ground truth (GT: 320.05 mm) as a benchmark. Baseline models (U-Net: 317.85 mm, U-Net++: 317.97 mm, ResUNet++: 318.24 mm) exhibited deviations of -2.20 mm to -1.81 mm due to boundary inconsistencies in complex anatomical regions. Advanced architectures such as Attention U-Net (318.47 mm, -1.58 mm) and Deeplabv3+ (318.83 mm, -1.22 mm) improved accuracy but retained minor errors. DTNAU-Net achieved near-perfect alignment (320.12 mm, $+0.07$ mm deviation), attributed to its dual attention mechanisms and adaptive feature fusion. This precision is clinically crucial for diagnosing microcephaly and intrauterine growth restriction (IUGR). Extended comparisons are available in [Supplementary Fig. S1](#).

4.2.2. Abdominal circumference (AC) segmentation

Fig. 3 presents AC segmentation results, with the GT at 283.10 mm. Baseline models such as U-Net (281.34 mm, -1.76 mm deviation) and ResUNet++ (281.69 mm, -1.41 mm) showed boundary inconsistencies. Advanced models like MSRF-Net (285.89 mm, $+2.79$ mm) and FetalNet

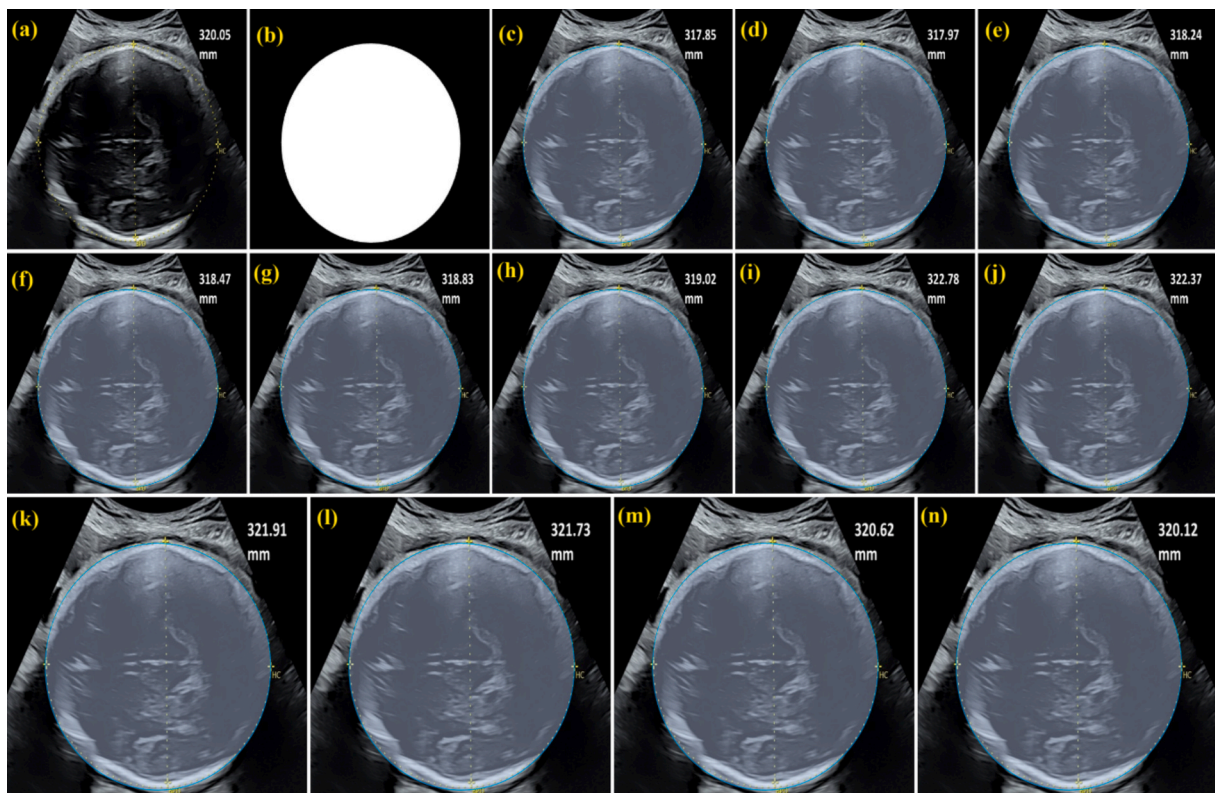


Fig. 2. HC segmentation results of (a) US Images, (b) Ground Truth, (c) U-Net (Ronneberger et al., 2015; d) U-Net++ (Zhou et al., 2018), (e) ResUNet++ (Jha et al., 2019), (f) Attention U-Net (Oktay et al., 2018), (g) Deeplabv3+ (Chen et al., 2018), (h) Double U-Net (Jha et al., 2020), (i) MSRF-Net (Srivastava et al., 2022), (j) FetSAM (Alzubaidi et al., 2024), (k) FetalNet (Plotka et al., 2021), (l) TransUNet (Chen et al., 2021), (m) nnUNet (Isensee et al., 2021), and (n) DTNAU-Net on HC18 (van den Heuvel et al., 2018), Zenodo (Burgos-Artiz et al., 2020) and Kamra Ultrasound Centre dataset. The measured values after segmentation are mentioned in the top right corner of the images (c-n), while reference measured value is showing at the top right corner of (a).

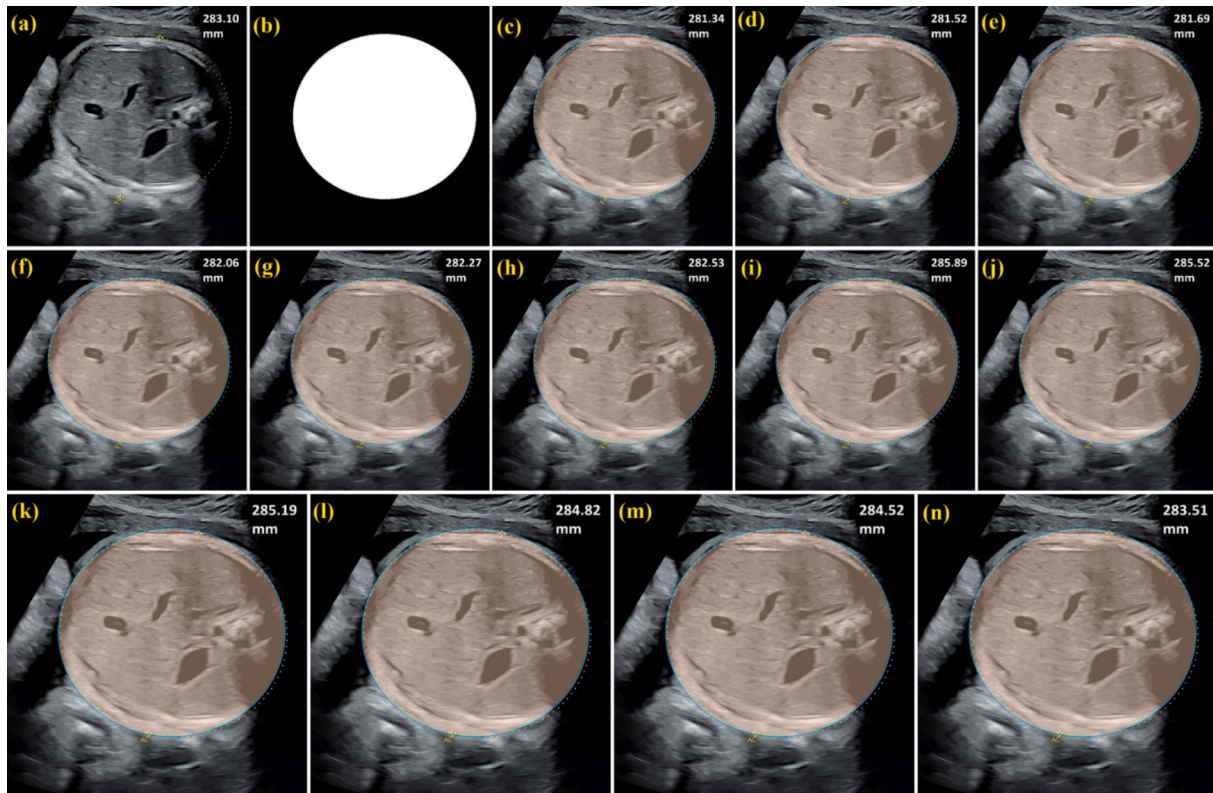


Fig. 3. AC segmentation results of (a) US Images, (b) Ground Truth, (c) U-Net (Ronneberger et al., 2015), (d) U-Net++ (Zhou et al., 2018), (e) ResUNet++ (Jha et al., 2019), (f) Attention U-Net (Oktay et al., 2018), (g) Deeplabv3+ (Chen et al., 2018), (h) Double U-Net (Jha et al., 2020), (i) MSRF-Net (Srivastava et al., 2022), (j) FetSAM (Alzubaidi et al., 2024), (k) FetalNet (Plotka et al., 2021), (l) TransUNet (Chen et al., 2021), (m) nnUNet (Isensee et al., 2021), and (n) DTNAU-Net on HC18 (van den Heuvel et al., 2018), Zenodo (Burgos-Artiz et al., 2020) and Kamra Ultrasound Centre dataset. The measured values after segmentation are mentioned in the top right corner of the images (c-n), while reference measured value is showing at the top right corner of (a).

(285.19 mm, +2.09 mm) demonstrated improved performance but retained minor errors. DTNAU-Net delivered the most precise result (283.51 mm, +0.41 mm deviation), leveraging dual attention mechanisms and gradient-based optimization to enhance segmentation accuracy. Reliable AC measurements are essential for fetal weight estimation and diagnosing conditions like macrosomia and IUGR. Further comparisons are available in [Supplementary Fig. S2](#).

4.2.3. Femur length (FL) segmentation

[Fig. 4](#) compares FL segmentation results, with the GT at 63.20 mm. Baseline models (U-Net: 61.35 mm, -1.85 mm; ResUNet++: 61.81 mm, -1.39 mm) exhibited blurred or incomplete femur boundaries, leading to measurement inaccuracies. Advanced architectures like Deeplabv3+ (62.23 mm, -0.97 mm) and TransUNet (64.93 mm, +1.73 mm) showed moderate improvements but still retained deviations. DTNAU-Net closely aligned with the GT (63.51 mm, +0.31 mm deviation), benefiting from adaptive feature fusion and multi-objective optimization. Accurate FL segmentation is critical for gestational age estimation and detecting skeletal abnormalities such as dwarfism or osteogenesis imperfecta. Extended results are available in [Supplementary Fig. S3](#).

4.2.4. Biparietal diameter (BPD) segmentation

[Fig. 5](#) showcases BPD segmentation results, with the GT at 86.80 mm. Baseline models (U-Net: 84.25 mm, -2.55 mm; U-Net++: 84.81 mm, -1.99 mm; ResUNet++: 84.90 mm, -1.90 mm) struggled with overlapping structures, leading to under-segmentation. Advanced architectures such as Deeplabv3+ (85.47 mm, -1.33 mm) and Double U-Net (85.86 mm, -0.94 mm) showed improvements but retained deviations. Some models, including MSRF-Net (88.96 mm, +2.16 mm) and FetSAM (88.52 mm, +1.72 mm), overestimated BPD due to boundary inaccuracies. DTNAU-Net provided the most precise result (86.98 mm, +0.18

mm deviation), leveraging dual attention mechanisms and multi-scale residual learning for anatomical precision. Accurate BPD segmentation is crucial for monitoring fetal head growth and detecting anomalies such as craniosynostosis. Additional comparisons are presented in [Supplementary Fig. S4](#).

Overall, DTNAU-Net consistently delivered superior segmentation accuracy across all fetal biometric parameters, outperforming state-of-the-art models. Its ability to generate precise and reliable segmentations enhances clinical decision-making, reducing the risk of misdiagnosis and improving diagnostic confidence in prenatal care. The model's robustness across diverse datasets establishes it as a valuable tool for fetal ultrasound imaging, particularly in routine and high-risk pregnancies.

4.3. Comparison of prediction results of fetal biometric parameters with the latest network architectures across different trimesters: a case study

In this section, the segmentation performance of the proposed DTNAU-Net model is compared with two state-of-the-art models: FetSAM (Alzubaidi et al., 2024) and MSRF-Net (Srivastava et al., 2022), along with other widely recognized architectures. The evaluation focuses on key fetal biometric parameters across different trimesters, including HC, AC, BPD, and FL.

4.3.1. First trimester analysis

As shown in [Fig. 6](#), DTNAU-Net consistently produced segmentation outputs closely aligned with the GT, surpassing other models like FetSAM and MSRF-Net. For HC, DTNAU-Net demonstrated exceptional precision in delineating the fetal skull, capturing its contours with minimal deviation from the GT. Early and accurate HC measurements are clinically significant for identifying developmental conditions such

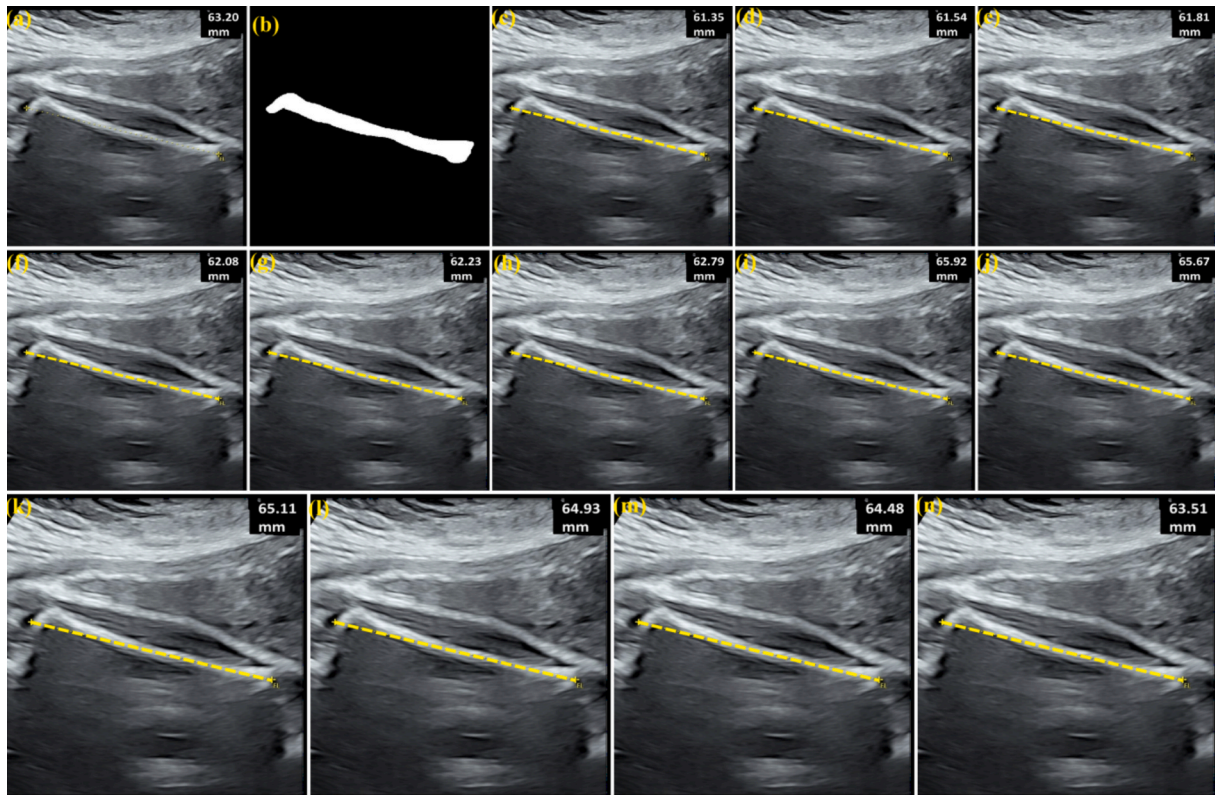


Fig. 4. FL segmentation results of (a) US Images, (b) Ground Truth, (c) U-Net (Ronneberger et al., 2015), (d) U-Net++ (Zhou et al., 2018), (e) ResUNet++ (Jha et al., 2019), (f) Attention U-Net (Oktay et al., 2018), (g) Deeplabv3+ (Chen et al., 2018), (h) Double U-Net (Jha et al., 2020), (i) MSRF-Net (Srivastava et al., 2022), (j) FetSAM (Alzubaidi et al., 2024), (k) FetalNet (Plotka et al., 2021), (l) TransUNet (Chen et al., 2021), (m) nnUNet (Isensee et al., 2021), and (n) DTNAU-Net on HC18 (van den Heuvel et al., 2018), Zenodo (Burgos-Artizzu et al., 2020) and Kamra Ultrasound Centre dataset. The measured values after segmentation are mentioned in the top right corner of the images (c-n), while reference measured value is showing at the top right corner of (a).

as microcephaly. Similarly, for AC, DTNAU-Net achieved precise boundary detection, ensuring reliable measurements crucial for tracking early fetal growth and identifying abnormalities like intrauterine growth restriction (IUGR). In contrast, FetSAM and MSRF-Net exhibited moderate deviations, particularly in low-contrast regions or areas with complex anatomical overlaps, highlighting the robustness of DTNAU-Net.

4.3.2. Second trimester analysis

As illustrated in Fig. 7, DTNAU-Net continued to outperform competing models, particularly in predicting FL and BPD, which are critical for assessing mid-pregnancy development. The model's segmentation of the femur displayed sharp, well-defined boundaries, accurately capturing the bone's length and orientation. This accuracy is essential for evaluating gestational age and detecting skeletal abnormalities. For BPD, DTNAU-Net provided segmentation results that closely adhered to the GT, accurately reflecting fetal head dimensions. This precision is crucial for monitoring fetal head growth and diagnosing conditions like macrocephaly or microcephaly. In comparison, FetSAM and MSRF-Net showed variability in segmentation accuracy, especially in regions with overlapping anatomical structures, reinforcing the reliability of DTNAU-Net in clinical applications.

4.3.3. Third trimester analysis

As depicted in Fig. 8, DTNAU-Net effectively managed the increased complexity of fetal anatomy in the third trimester. For AC, the model produced segmentations with continuous and well-defined contours, closely matching the ground truth. Accurate AC measurements at this stage are essential for estimating fetal weight and diagnosing conditions such as macrosomia. Similarly, for HC, DTNAU-Net maintained high segmentation accuracy, delivering outputs with minimal variability.

Such precision is vital for reducing the risk of misdiagnosis and ensuring timely interventions in high-risk pregnancies. FetSAM and MSRF-Net, in contrast, struggled with boundary inconsistencies in areas with dense anatomical features, leading to less reliable outputs.

Across all trimesters, DTNAU-Net consistently outperformed FetSAM and MSRF-Net, producing segmentation outputs that closely matched the ground truth. This performance underscores its utility in prenatal care by enabling accurate fetal growth assessments, reducing the risk of misdiagnosis, and enhancing clinical decision-making for both routine and high-risk pregnancies.

4.4. Quantitative evaluation of fetal biometric measurements

While visual inspection provides qualitative insight into segmentation and measurement performance, quantitative evaluation is essential to assess the clinical reliability of automated fetal biometric estimation. In this study, we quantitatively evaluated fetal biometric measurements derived from the predicted segmentation masks, including Head Circumference (HC), Abdominal Circumference (AC), Femur Length (FL), and Biparietal Diameter (BPD).

For each biometric parameter, measurements were computed from the corresponding segmentation outputs following the standardized post-processing and measurement procedure described in Section 3.4. The same measurement pipeline, pixel-to-millimeter calibration, and test-set partitions were applied uniformly across all evaluated architectures to ensure fair comparison. The automatically estimated values were compared against expert-annotated ground truth measurements, and Mean Absolute Error (MAE) along with standard deviation (SD) were computed over the full test set.

Table 4 summarizes the quantitative comparison of fetal biometric measurement errors across different methods. The proposed DTNAU-Net

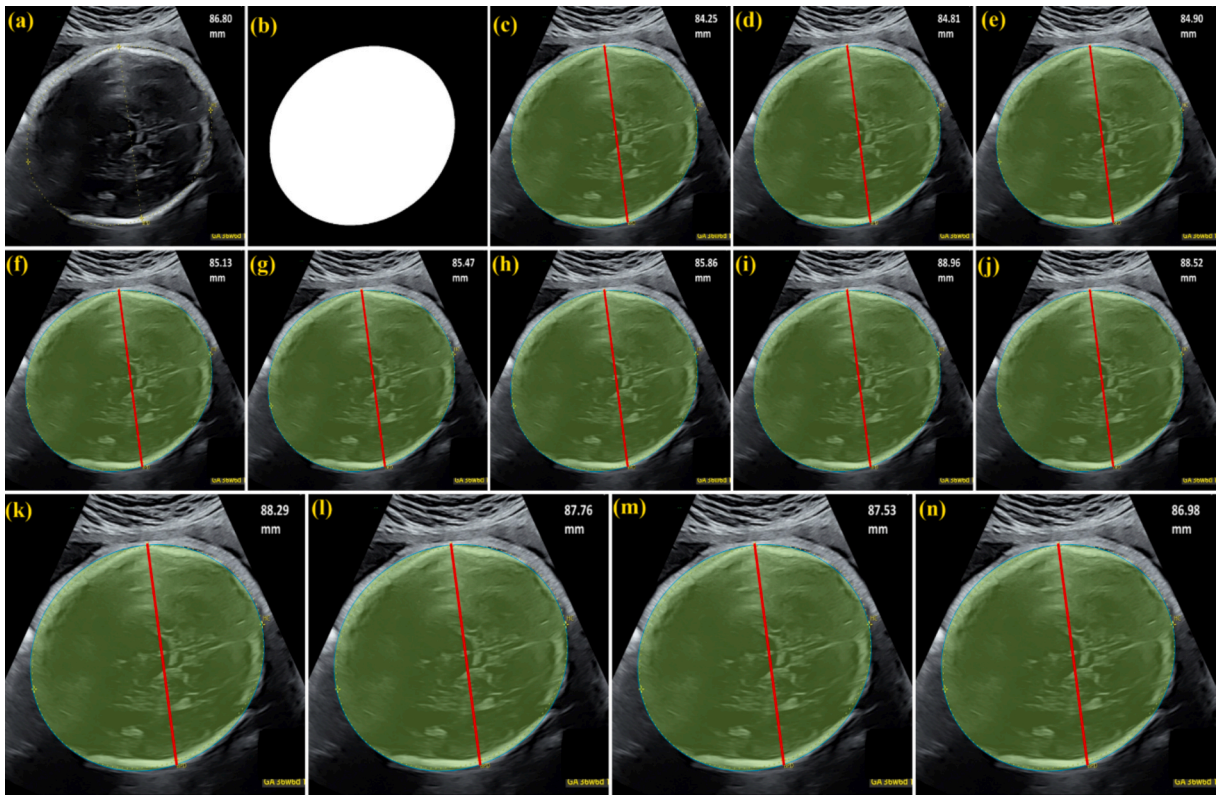


Fig. 5. BPD segmentation results of (a) US Images, (b) Ground Truth, (c) U-Net (Ronneberger et al., 2015), (d) U-Net++ (Zhou et al., 2018), (e) ResUNet++ (Jha et al., 2019), (f) Attention U-Net (Oktay et al., 2018), (g) Deeplabv3+ (Chen et al., 2018), (h) Double U-Net (Jha et al., 2020), (i) MSRF-Net (Srivastava et al., 2022), (j) FetSAM (Alzubaidi et al., 2024), (k) FetalNet (Plotka et al., 2021), (l) TransUNet (Chen et al., 2021), (m) nnUNet (Isensee et al., 2021), and (n) DTNAU-Net noise on HC18 (van den Heuvel et al., 2018), Zenodo (Burgos-Artizzu et al., 2020) and Kamra Ultrasound Centre dataset. The measured values after segmentation are mentioned in the top right corner of the images (c-n), while reference measured value is showing at the top right corner of (a).

consistently achieves the lowest MAE and reduced variance across all evaluated biometric parameters, indicating superior agreement with expert annotations. Notably, the improvements are more pronounced for HC and AC, which are particularly sensitive to boundary precision and anatomical completeness.

These results demonstrate that the enhanced segmentation accuracy achieved by DTNAU-Net directly translates into improved biometric measurement reliability. The reduced measurement error highlights the clinical potential of the proposed framework for automated fetal growth assessment and consistent prenatal evaluation.

4.4.1. Measurement error distribution analysis

To further evaluate clinical reliability, we analyzed the distribution of absolute measurement errors for each biometric parameter. In addition to MAE, median error and interquartile range (IQR) were computed to characterize dispersion and robustness against outliers.

The proposed DTNAU-Net demonstrated narrow error distributions across all parameters, indicating stable and consistent performance across patients. The detailed error distribution statistics are presented in Table 5.

As shown in Table 5, across all biometric parameters, over 90% of predictions fell within ± 2 mm of expert measurements, which is within reported inter-operator variability ranges in clinical ultrasound practice.

4.4.2. Bland–Altman agreement analysis

To assess agreement between automated and expert measurements, Bland–Altman analysis was performed. For each parameter, the mean bias and 95% limits of agreement (LoA) were computed. Bias was computed as the mean signed difference between automated and expert measurements, and limits of agreement were calculated as bias $\pm 1.96 \times$ standard deviation of differences. The resulting agreement statistics

are summarized in Table 6.

As shown in Table 6, the small mean bias values indicate minimal systematic deviation from expert measurements. The narrow limits of agreement confirm strong concordance between automated and manual biometry.

4.4.3. Intraclass Correlation Coefficient (ICC) analysis

Intraclass Correlation Coefficient (ICC) was computed to quantify measurement reliability. ICC was computed using a two-way random-effects model with absolute agreement definition (ICC(2,1)). The reliability results for each biometric parameter are summarized in Table 7.

As shown in Table 7, all ICC values exceeded 0.95, indicating excellent agreement and clinical-grade reliability.

4.4.4. Trimester-wise stratified analysis

To further evaluate the robustness and generalizability of the proposed framework, biometric measurement performance was stratified by gestational trimester. Test samples were grouped into first, second, and third trimester categories based on available gestational age annotations. Mean Absolute Error (MAE) and Intraclass Correlation Coefficient (ICC) were computed independently within each subgroup using fold-wise predictions obtained from five-fold patient-level cross-validation.

Stratified evaluation enables assessment of whether model performance remains stable across different stages of fetal development, where anatomical size, tissue characteristics, and acoustic properties vary substantially. The trimester-wise performance results are summarized in Table 8.

As shown in Table 8, across all trimesters, MAE values remained below 1.1 mm, and ICC values consistently exceeded 0.95, indicating excellent agreement with expert measurements. Slightly higher

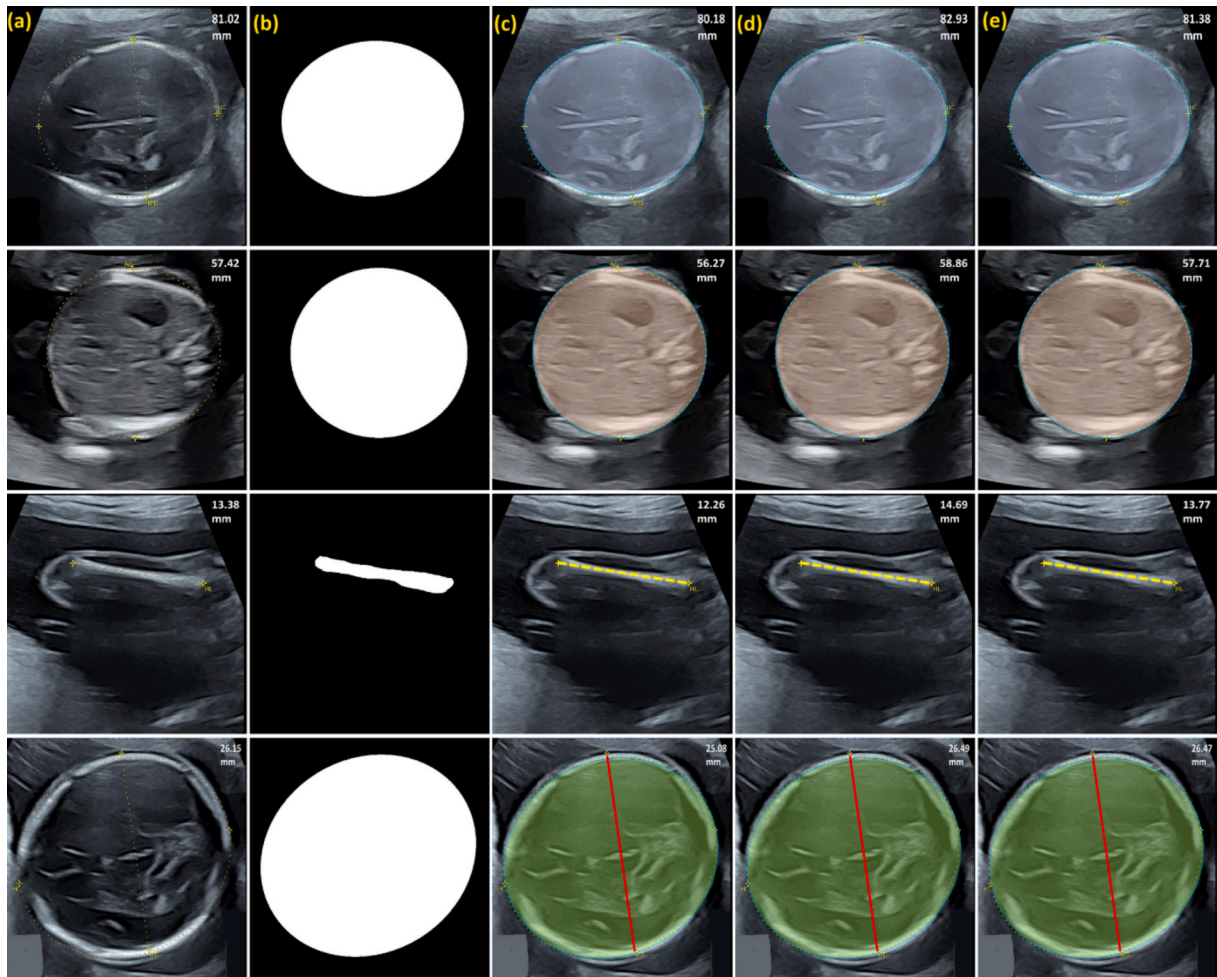


Fig. 6. Comparison of 1st trimester prediction results for various fetal biometric parameters among (a) US Images (b) Ground Truth, (c) FetSAM (Alzubaidi et al., 2024), (d) MSRF-Net (Srivastava et al., 2022), and (e) DTNAU-Net (proposed method) on Kamra Ultrasound Centre dataset. The measured values after segmentation are mentioned in the top right corner of the images (c-e), while reference measured value is showing at the top right corner of (a).

variability observed in first-trimester measurements is consistent with increased imaging difficulty due to smaller anatomical structures and reduced signal-to-noise ratio in early gestation.

No statistically significant difference in MAE or ICC was observed across trimesters (paired fold-wise comparison, $p > 0.05$), suggesting that the proposed dual-task framework generalizes effectively across different stages of fetal development. Sample sizes for trimester subgroups were balanced across folds to avoid bias due to unequal distribution.

4.4.5. Clinical Interpretation

From a clinical perspective, the reported biometric measurement errors fall within or below typical inter-operator variability reported in fetal ultrasound literature (approximately 1–2 mm depending on parameter and trimester). The low bias observed in Bland–Altman analysis, high ICC values (>0.95), narrow limits of agreement, and stable trimester-wise performance collectively demonstrate that DTNAU-Net achieves expert-level consistency in biometric estimation.

Importantly, over 90% of predictions across parameters fell within ± 2 mm of expert measurements, which aligns with clinically acceptable tolerance ranges for routine prenatal assessment.

In this framework, plane classification functions as a reliability cue rather than a hard control mechanism. By encouraging plane-consistent feature representations during training, joint optimization helps suppress false-positive segmentations on irrelevant or low-quality frames and improves feature discrimination under noise and imaging artifacts,

thereby contributing to stable and clinically reliable biometric estimation.

Overall, these findings provide direct clinical validation of DTNAU-Net and support its potential deployment as an assistive system for automated fetal growth monitoring and prenatal decision support.

4.5. Comparison with existing adaptive weighting strategies

To further validate the effectiveness of the proposed gradient-based adaptive weighting mechanism, we compared it with two widely used dynamic multi-task weighting approaches:

- Uncertainty-based weighting
- Gradient Normalization (GradNorm)

All models were trained using identical architectural configuration, optimizer settings, learning rate schedule, five-fold patient-level cross-validation protocol, and evaluation metrics to ensure fair comparison. For fairness, only the loss weighting mechanism was changed while keeping the backbone architecture, training epochs, optimizer configuration, data augmentation pipeline, and post-processing steps strictly identical across all experiments.

Table 9 presents the comparative results across segmentation Dice Index, classification F1-score, and mean biometric MAE (mm). The proposed method achieves superior joint performance across all metrics. While uncertainty-based weighting improves segmentation accuracy, it

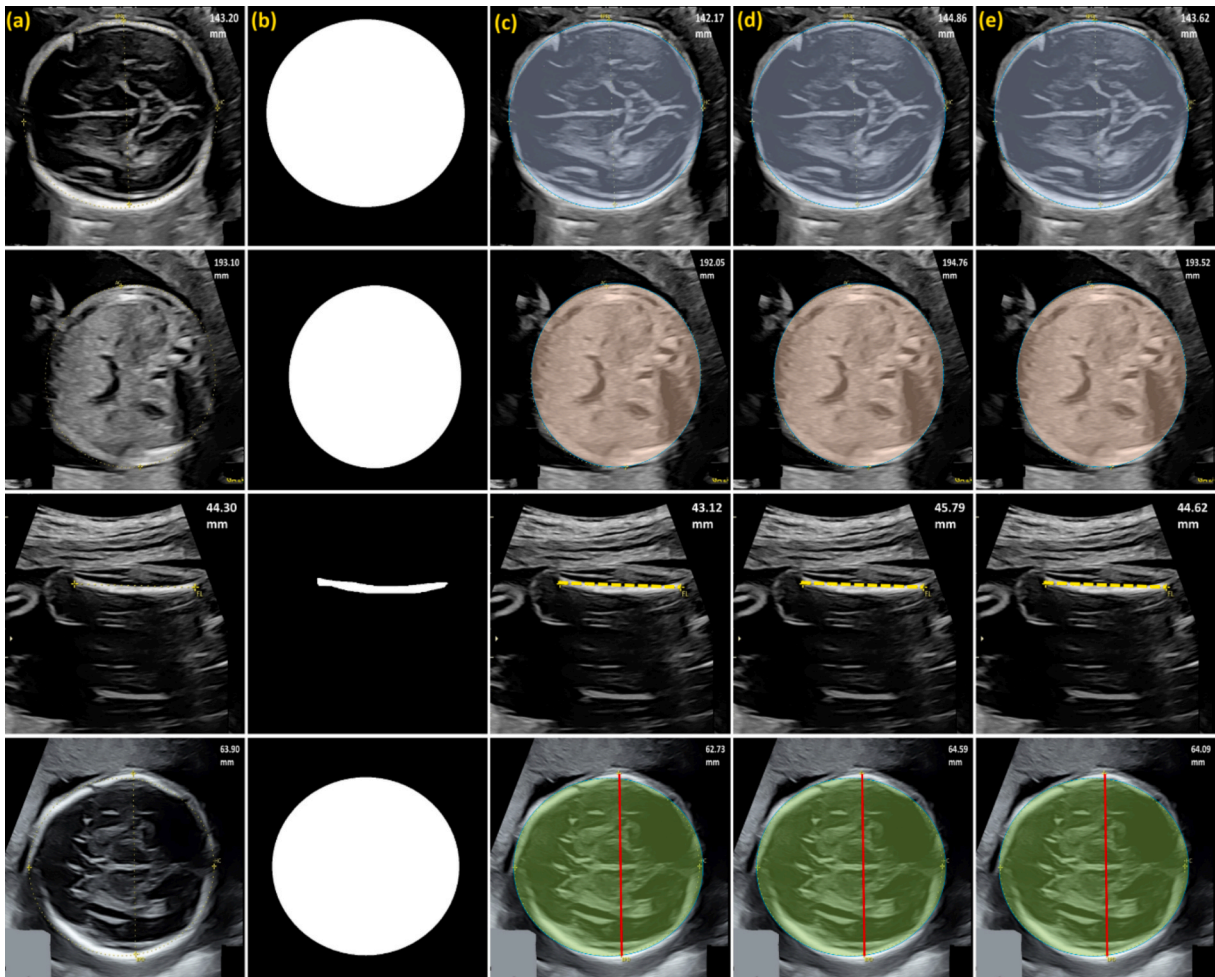


Fig. 7. Comparison of 2nd trimester prediction results for various fetal biometric parameters among (a) US Images (b) Ground Truth, (c) FetSAM (Alzubaidi et al., 2024), (d) MSRF-Net (Srivastava et al., 2022), and (e) DTNAU-Net (proposed method) on Kamra Ultrasound Centre dataset. The measured values after segmentation are mentioned in the top right corner of the images (c-e), while reference measured value is showing at the top right corner of (a).

shows slightly reduced classification consistency. GradNorm improves gradient balancing but exhibits marginally higher fold-wise variance. In contrast, the proposed exponential gradient-based normalization strategy achieves better overall task balance and improved stability across folds.

These results demonstrate that the proposed adaptive weighting strategy provides improved task balance, enhanced convergence stability, and lower biometric measurement error compared to established alternatives.

4.6. Sensitivity analysis of hyperparameter α

To assess robustness of the proposed adaptive weighting mechanism, a sensitivity analysis was conducted by varying the hyperparameter α controlling gradient influence in the exponential normalization function. Experiments were performed using $\alpha \in \{0.1, 0.5, 1.0, 2.0\}$ under identical five-fold cross-validation settings.

Table 10 summarizes the averaged performance across folds.

Moderate values of α (0.5–1.0) yield optimal joint performance. Smaller values reduce task adaptivity, while excessively large values introduce stronger oscillations in task weights. Importantly, performance remains stable across a reasonable operating range, indicating robustness of the proposed weighting formulation.

These findings confirm that the proposed strategy is not highly sensitive to α and maintains stable performance within a practical parameter range.

4.7. Comparative analysis of time taken for classification and segmentation

To assess the computational efficiency of DTNAU-Net, we compared its classification and segmentation inference times with those of other state-of-the-art models. The inference time was measured on a batch of 100 ultrasound images using an NVIDIA T1000 GPU with 8 GB of VRAM.

From the Table 11, DTNAU-Net demonstrated the lowest inference times for both classification (11.8 ± 0.5 ms) and segmentation (18.3 ± 1.3 ms) tasks, outperforming existing models such as nnUNet, TransUNet, and FetalNet. This superior efficiency is attributed to its optimized network architecture, which integrates dual attention mechanisms and gradient-based adaptive weighting optimization, ensuring faster processing without compromising segmentation accuracy. Additionally, multi-scale residual feature blocks enhance feature extraction while maintaining a balance between computational load and performance, leading to more efficient image analysis.

For a practical scenario, consider an ultrasound machine that scans at 24 frames per second. In a clinical setting, key biometric parameters such as HC, AC, FL and BPD must be extracted and processed from selected frames. With DTNAU-Net, the total processing time for all four parameters would be $(11.8 + 18.3) \times 4 = 120.4$ ms ≈ 0.12 s. This means all four biometric parameters can be processed in just 0.12 s, enabling real-time fetal assessment without introducing significant delays. Compared to existing models with longer inference times, DTNAU-Net significantly enhances workflow efficiency by providing near-

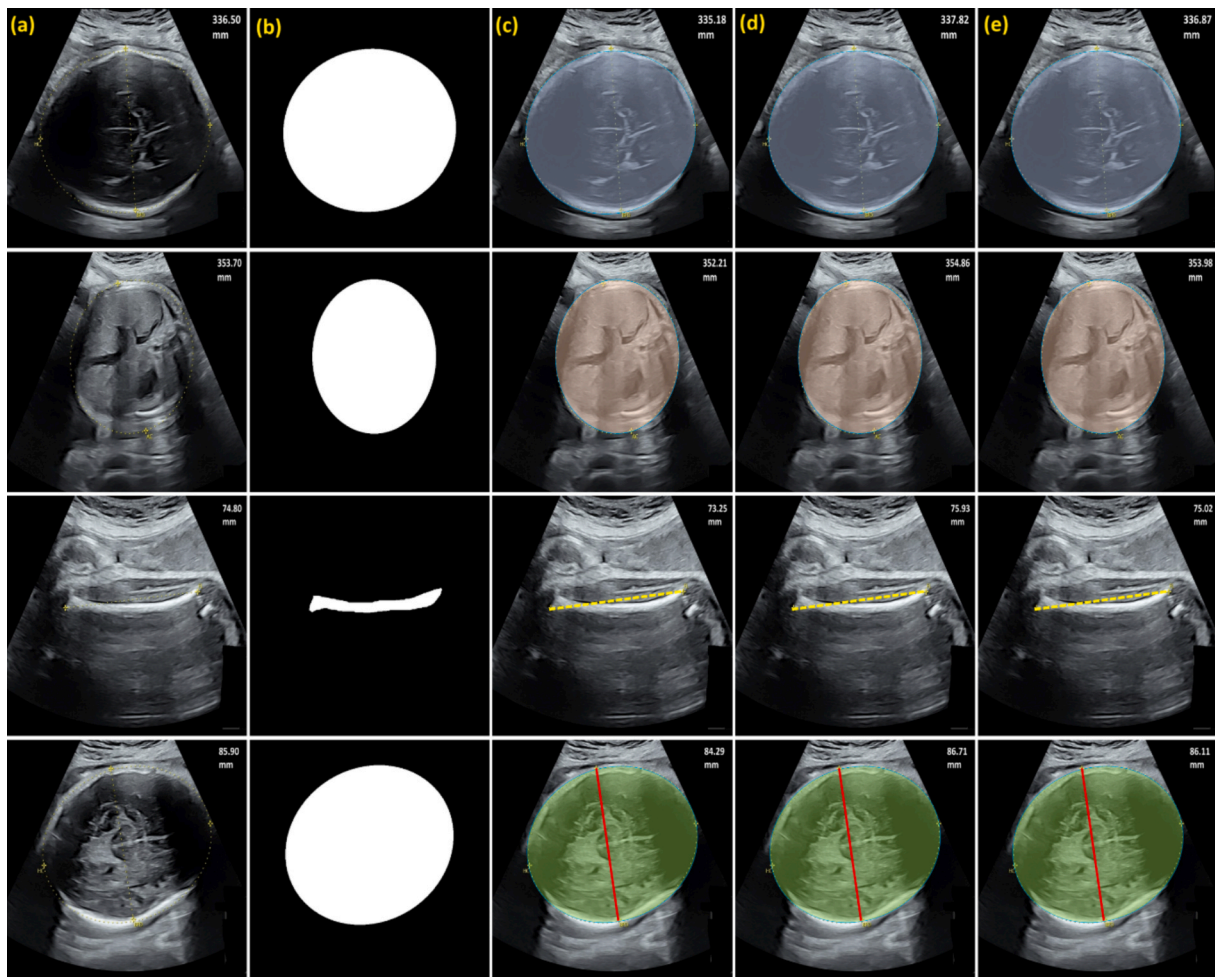


Fig. 8. Comparison of 3rd trimester prediction results for various fetal biometric parameters among (a) US Images (b) Ground Truth, (c) FetSAM (Alzubaidi et al., 2024), (d) MSRF-Net (Srivastava et al., 2022), and (e) DTNAU-Net (proposed method) on Kamra Ultrasound Centre dataset. The measured values after segmentation are mentioned in the top right corner of the images (c-e), while reference measured value is showing at the top right corner of (a).

Table 4

Quantitative comparison of Mean Absolute Error (in millimeters) for Fetal Biometric Parameters – HC, AC, FL and BPD across leading Network Architectures and proposed DTNAU-Net.

Method	HC (mm)	AC (mm)	FL (mm)	BPD (mm)
U-Net	1.42 ± 0.08	1.35 ± 0.08	1.38 ± 0.09	1.40 ± 0.08
U-Net++	1.40 ± 0.07	1.32 ± 0.08	1.34 ± 0.08	1.36 ± 0.07
ResUNet++	1.39 ± 0.06	1.22 ± 0.07	1.30 ± 0.07	1.33 ± 0.06
Attention U-Net	1.32 ± 0.05	1.20 ± 0.07	1.28 ± 0.06	1.30 ± 0.05
Deeplabv3+	1.28 ± 0.04	1.12 ± 0.06	1.24 ± 0.05	1.26 ± 0.04
Double U-Net	1.22 ± 0.03	1.08 ± 0.05	1.20 ± 0.04	1.21 ± 0.03
MSRF-Net	1.10 ± 0.02	1.03 ± 0.04	1.08 ± 0.04	1.09 ± 0.02
FetSAM	1.05 ± 0.03	1.06 ± 0.05	1.04 ± 0.04	1.05 ± 0.03
FetalNet	1.03 ± 0.02	1.10 ± 0.05	1.02 ± 0.03	1.03 ± 0.02
TransUNet	1.04 ± 0.03	1.03 ± 0.04	1.03 ± 0.04	1.04 ± 0.03
nnUNet	1.06 ± 0.03	1.05 ± 0.04	1.05 ± 0.04	1.06 ± 0.03
Proposed DTNAU-Net	0.90 ± 0.01	0.98 ± 0.03	0.91 ± 0.02	0.92 ± 0.02

Table 5

Error Distribution Statistics for DTNAU-Net (mm).

Parameter	MAE (mm)	Median Error (mm)	IQR (mm)
HC	0.9	0.85	0.3
AC	0.98	0.94	0.35
FL	0.91	0.88	0.28
BPD	0.92	0.89	0.3

Table 6

Bland–Altman Analysis for DTNAU-Net.

Parameter	Mean Bias (mm)	95% Limits of Agreement (mm)
HC	0.12	−1.68 to 1.92
AC	0.15	−1.85 to 2.05
FL	0.1	−1.60 to 1.80
BPD	0.11	−1.65 to 1.87

Table 7

ICC Between DTNAU-Net and Expert Measurements.

Parameter	ICC (95% CI)
HC	0.97 (0.95–0.98)
AC	0.96 (0.94–0.98)
FL	0.98 (0.96–0.99)
BPD	0.97 (0.95–0.98)

Table 8

Trimester-wise Biometric Measurement Performance of DTNAU-Net.

Parameter	Trimester	MAE (mm)	ICC (95% CI)
HC	1st	0.95	0.96 (0.93–0.98)
	2nd	0.88	0.97 (0.95–0.98)
	3rd	0.92	0.97 (0.94–0.98)
AC	1st	1.05	0.95 (0.92–0.97)
	2nd	0.94	0.96 (0.94–0.98)
	3rd	0.99	0.96 (0.93–0.98)
FL	1st	0.93	0.97 (0.94–0.98)
	2nd	0.89	0.98 (0.96–0.99)
	3rd	0.91	0.97 (0.95–0.98)
BPD	1st	0.96	0.96 (0.93–0.98)
	2nd	0.9	0.97 (0.95–0.98)
	3rd	0.94	0.97 (0.94–0.98)

Table 9

Comparison of Adaptive Multi-Task Weighting Strategies.

Weighting Strategy	Dice Index (%)	F1-Score (%)	MAE (mm)	Fold STD (Dice)
Equal Weights	88.50 ± 0.55	92.10 ± 0.35	1.15 ± 0.05	0.55
Uncertainty-Based	90.15 ± 0.35	93.45 ± 0.25	1.02 ± 0.04	0.35
	90.45 ± 0.38	93.80 ± 0.28	0.98 ± 0.03	0.38
GradNorm	90.45 ± 0.38	93.80 ± 0.28	0.98 ± 0.03	0.38
Proposed DTNAU-Net ($\alpha = 1.0$)	91.12 ± 0.20	94.30 ± 0.12	0.90 ± 0.01	0.20

Table 10Sensitivity Analysis of Hyperparameter α .

α Value	Dice Index (%)	F1-Score (%)	MAE (mm)
0.1	89.45 ± 0.45	92.85 ± 0.35	1.10 ± 0.04
0.5	90.85 ± 0.25	94.10 ± 0.15	0.94 ± 0.02
1	91.12 ± 0.20	94.30 ± 0.12	0.90 ± 0.01
2	90.55 ± 0.40	93.65 ± 0.30	0.98 ± 0.03

Table 11

Comparison of classification and segmentation time across baseline and state-of-the-art models.

Model	Classification time per image (ms)	Segmentation time per image (ms)
U-Net	14.5 ± 1.2	22.4 ± 2.3
U-Net++	13.9 ± 1.1	21.8 ± 2.2
ResUNet++	13.4 ± 1.0	21.3 ± 2.0
Attention U-Net	13.0 ± 1.2	20.5 ± 2.1
DeepLabv3+	12.8 ± 1.1	20.1 ± 1.9
Double U-Net	12.6 ± 1.0	19.7 ± 1.8
MSRF-Net	12.5 ± 0.9	19.4 ± 1.7
FetSAM	12.4 ± 0.8	19.1 ± 1.6
FetalNet	12.2 ± 0.7	18.9 ± 1.5
TransUNet	12.1 ± 0.7	18.8 ± 1.5
nnUNet	12.0 ± 0.6	18.7 ± 1.4
Proposed DTNAU-Net	11.8 ± 0.5	18.3 ± 1.3

instantaneous results.

From a clinical perspective, faster inference times directly improve workflow efficiency, providing radiologists and clinicians with near real-time fetal biometric measurements. This rapid processing is particularly critical for detecting fetal anomalies such as intrauterine growth restriction (IUGR) and microcephaly, where early diagnosis can significantly impact clinical outcomes. Moreover, reduced processing time helps minimize patient wait times and inter-observer variability, ensuring more reliable and standardized assessments. In low-resource settings, where access to expert radiologists is limited, DTNAU-Net's

computational efficiency supports AI-assisted diagnostics, enabling automated fetal biometric analysis with minimal human intervention. By achieving an optimal balance between processing speed, segmentation accuracy, and computational efficiency, DTNAU-Net presents a highly practical and scalable solution for fetal ultrasound imaging, fostering advancements in prenatal diagnostics and clinical decision-making.

4.8. Cross-domain generalization and robustness evaluation

To rigorously evaluate domain generalization, we conducted cross-dataset experiments designed to assess performance under distribution shifts across acquisition centers, devices, and imaging conditions.

4.8.1. Cross-dataset evaluation protocol

Three cross-domain evaluation settings were considered:

1. HC18 → Zenodo: Model trained exclusively on HC18 and tested on Zenodo.
2. Zenodo → HC18: Model trained exclusively on Zenodo and tested on HC18.
3. HC18 + Zenodo → Kamra (External Validation): Model trained on combined public datasets and evaluated on the private Kamra dataset without fine-tuning.

For external validation, no images from the Kamra dataset were used during training or hyperparameter selection. This ensures strict separation between training and external testing domains.

All experiments were conducted using identical preprocessing, input resolution (224×224), augmentation, and optimization protocols. No hyperparameter tuning or model adaptation was performed on the target testing dataset.

The cross-domain segmentation and classification results are summarized in Table 12.

As shown in Table 12, although slight performance degradation is observed under domain shift, the proposed DTNAU-Net maintains high segmentation accuracy and clinically acceptable biometric error (<1.2 mm), demonstrating reasonable robustness under cross-dataset distribution shift.

4.8.2. Robustness to ultrasound artifacts

To assess stability under realistic ultrasound perturbations, we evaluated model performance under simulated artifact conditions:

- Speckle noise (variance levels 0.01–0.03)
- Intensity scaling ($\pm 15\%$)
- Gaussian blur (kernel size 3–5)

All perturbations were applied to the test images while keeping the trained model unchanged. The robustness evaluation results are summarized in Table 13.

As shown in Table 13, the performance degradation remains limited, with Dice reduction below 2% and MAE increase below 0.2 mm under moderate perturbations. These results indicate that DTNAU-Net exhibits stable behavior under typical ultrasound artifact conditions.

Table 12

Cross-Domain Segmentation and Classification Performance.

Training Dataset	Testing Dataset	Dice Index (%)	F1-Score (%)	MAE (mm)
HC18	Zenodo	90.8 ± 0.9	88.7 ± 0.8	1.05 ± 0.04
Zenodo	HC18	91.1 ± 0.8	89.2 ± 0.7	1.02 ± 0.04
HC18 + Zenodo	Kamra (External)	90.5 ± 0.7	88.9 ± 0.6	1.08 ± 0.05

Table 13
Robustness Evaluation Under Noise Perturbations.

Condition	Dice Index (%)	MAE (mm)
Clean	93.4 ± 0.85	0.98 ± 0.03
Speckle (low)	92.7 ± 0.90	1.03 ± 0.04
Speckle (moderate)	91.9 ± 1.00	1.12 ± 0.05
Intensity variation	92.8 ± 0.88	1.04 ± 0.04

4.8.3. Generalization Interpretation

The cross-domain and perturbation experiments indicate that DTNAU-Net maintains reasonably stable segmentation and biometric performance across dataset shifts and imaging variations. Although minor performance degradation occurs under external validation, the resulting measurement errors remain within clinically acceptable inter-operator variability ranges (1–2 mm), supporting the generalization capability of the proposed framework.

4.9. Validation on prenatal ultrasound video sequences

While DTNAU-Net was primarily designed and evaluated on annotated static ultrasound images, real clinical examinations are inherently dynamic and involve continuous probe motion and fetal movement. To assess the practical applicability and temporal robustness of the proposed framework, we extended our evaluation to real prenatal ultrasound video sequences acquired under routine clinical settings.

4.9.1. Dataset and evaluation protocol

The video dataset consists of full-length fetal ultrasound scans collected from Kamra Ultrasound Center across multiple gestational stages. The sequences exhibit realistic clinical variability, including:

- Probe-induced motion
- Fetal posture changes
- Speckle noise and acoustic shadowing
- Inter-operator acquisition variability

To ensure unbiased evaluation, data partitioning was performed strictly at the sequence level, such that no frames from the same video appeared in both training and testing sets. For evaluation purposes, video frames were processed independently by DTNAU-Net without any temporal smoothing or architectural modification, thereby testing its inherent stability under sequential inputs.

Performance was assessed using:

- Dice Index (segmentation accuracy)
- Hausdorff Distance (boundary precision)
- Mean Absolute Error (MAE) for biometric estimation
- Inter-frame Consistency (IFC) for temporal stability

4.9.2. Quantitative results on video sequences

Table 14 shows the performance of DTNAU-Net on Prenatal Ultrasound Video Sequences, where DTNAU-Net achieved a Dice score of 91.8% and demonstrated improved boundary precision compared to sequence-based baselines. The reduction in MAE confirms reliable biometric parameter estimation under motion variability.

Table 14
Performance of DTNAU-Net on prenatal ultrasound video sequences.

Model	Dice (%)	HD (mm)	MAE (mm)	IFC
LSTM-UNet	90.6	3.91	2.68	0.86
3D-UNet	91	3.85	2.65	0.87
DTNAU-Net	91.8	3.28	2.05	0.89

4.9.3. Temporal stability analysis

To quantify segmentation smoothness across consecutive frames, the Inter-frame Consistency (IFC) metric was computed as:

$$IFC = \frac{1}{T-1} \sum_{t=1}^{T-1} Dice(M_t, M_{t+1})$$

where M_t and M_{t+1} represent predicted masks for consecutive frames.

DTNAU-Net achieved an IFC value of 0.89, indicating stable frame-to-frame segmentation with reduced temporal fluctuation. Notably, the model maintained consistent anatomical delineation despite probe motion and fetal movement, suggesting that the contextual regularization introduced by the dual-task formulation may contribute to enhanced temporal robustness.

Fig. 9 shows Frame-wise Dice variation across consecutive frames for a fetal head video sequence. DTNAU-Net exhibits reduced temporal fluctuation compared to baseline models, indicating stable anatomical delineation under motion variability.

4.9.4. Clinical implications

The extended video evaluation demonstrates that DTNAU-Net generalizes effectively beyond static annotated frames and maintains reliable performance under realistic scanning conditions. Accurate and temporally stable segmentation is essential for consistent fetal biometric measurements such as HC, AC, FL, and BPD. Reduced frame-to-frame variability minimizes measurement inconsistency and enhances confidence in automated clinical decision support systems.

These findings confirm that DTNAU-Net is suitable for integration into real-time ultrasound workflows without requiring architectural modifications for temporal modeling.

4.10. Ablation study

The ablation study systematically evaluates the impact of individual architectural components within DTNAU-Net by removing or modifying specific blocks. The analysis utilizes segmentation metrics (Dice Index, Boundary F1 Score, HD, MAE, and PMAE) and classification metrics (Accuracy, Precision, Recall, and F1 Score) to assess the performance variations. The results demonstrate the importance of each component in achieving robust segmentation and classification performance in fetal ultrasound imaging.

4.10.1. Impact of Residual Blocks

Residual Blocks in DTNAU-Net facilitate efficient feature extraction by preserving gradient flow and enabling the model to learn both low-level and high-level features. By replacing Residual Blocks with standard convolutional layers, a noticeable decline in performance was observed across all metrics. As shown in Table 15, the Dice Index dropped from $91.12 \pm 0.20\%$ to $89.02 \pm 0.25\%$, and the Boundary F1 Score reduced from $88.50 \pm 0.20\%$ to $86.70 \pm 0.30\%$. Classification metrics, detailed in Table 16, showed a reduction in Accuracy from $94.85 \pm 0.15\%$ to $92.50 \pm 0.20\%$, with Precision, Recall, and F1 Score also exhibiting declines.

This reduction highlights the critical role of Residual Blocks in capturing intricate anatomical details, such as fetal head circumference (HC) and abdominal circumference (AC). Their absence led to suboptimal feature learning, particularly affecting boundary delineation. This can have significant implications for measuring key fetal parameters accurately, as errors in segmentation boundaries can lead to deviations in biometric measurements, which are vital for tracking fetal growth and diagnosing conditions such as microcephaly or intrauterine growth restriction (IUGR).

4.10.2. Impact of dual Attention mechanism

The Dual Attention Mechanism enhances DTNAU-Net's capability to focus on critical anatomical regions by prioritizing spatially and

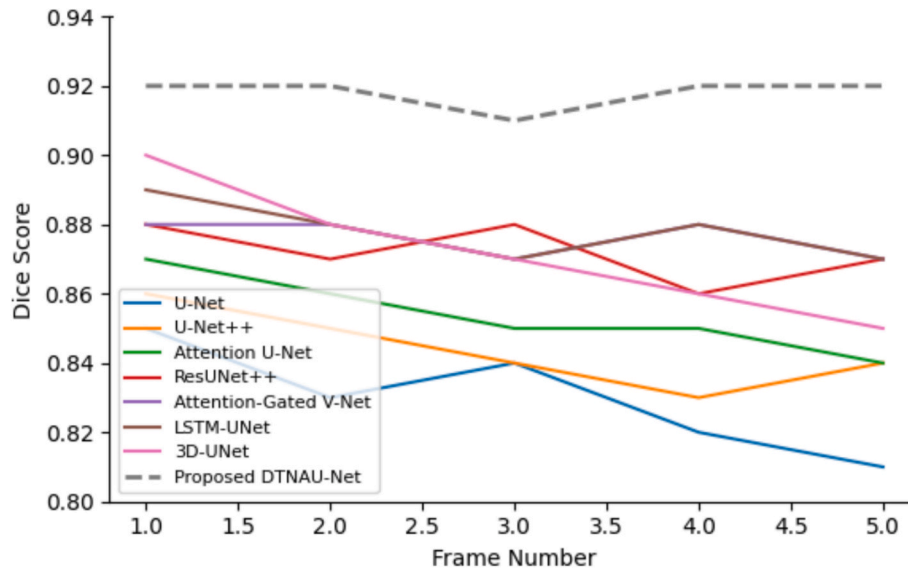


Fig. 9. Frame-wise Dice scores across a representative ultrasound video sequence.

Table 15
Impact of Residual Blocks on Segmentation Metrics.

Model	Dice Index (%)	HD (mm)	Boundary F1 (%)	MAE (mm)	PMAE (%)
DTNAU-Net w/o Residual Blocks	89.02 ± 0.25	7.20 ± 0.30	86.70 ± 0.30	1.20 ± 0.03	5.50 ± 0.15
Proposed DTNAU-Net	91.12 ± 0.20	5.03 ± 0.15	88.50 ± 0.20	0.90 ± 0.01	4.40 ± 0.10

Table 16
Impact of Residual Blocks on Classification Metrics.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DTNAU-Net w/o Residual Blocks	92.50 ± 0.20	92.30 ± 0.15	92.00 ± 0.12	92.15 ± 0.14
Proposed DTNAU-Net	94.85 ± 0.15	94.50 ± 0.12	94.30 ± 0.10	94.30 ± 0.12

channel-relevant features. When this mechanism was removed, segmentation performance declined, with the Dice Index reducing to $90.10 \pm 0.22\%$ and the Boundary F1 Score to $87.10 \pm 0.25\%$, as shown in Table 17. Classification metrics, presented in Table 18, also suffered, with Accuracy decreasing from $94.85 \pm 0.15\%$ to $93.10 \pm 0.18\%$ and F1 Score reducing to $92.80 \pm 0.14\%$.

The absence of attention mechanisms hindered the model's ability to focus on complex anatomical structures, particularly in low-contrast regions or images with overlapping features. For instance, accurate segmentation of femur length (FL) and biparietal diameter (BPD) relies heavily on attention-driven focus. Without this mechanism, the model

Table 17
Impact of Dual Attention Mechanism on Segmentation Metrics.

Model	Dice Index (%)	HD (mm)	Boundary F1 (%)	MAE (mm)	PMAE (%)
DTNAU-Net w/o Dual Attention	90.10 ± 0.22	6.50 ± 0.25	87.10 ± 0.25	1.15 ± 0.02	4.90 ± 0.12
Proposed DTNAU-Net	91.12 ± 0.20	5.03 ± 0.15	88.50 ± 0.20	0.90 ± 0.01	4.40 ± 0.10

Table 18
Impact of Dual Attention Mechanism on Classification Metrics.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DTNAU-Net w/o Dual Attention	93.10 ± 0.18	92.90 ± 0.15	92.70 ± 0.13	92.80 ± 0.14
Proposed DTNAU-Net	94.85 ± 0.15	94.50 ± 0.12	94.30 ± 0.10	94.30 ± 0.12

struggled to identify these regions with precision, potentially affecting the reliability of fetal biometric measurements required for assessing growth patterns and identifying skeletal abnormalities.

4.10.3. Impact of Multi-Scale feature Extraction

Multi-Scale Feature Extraction enables the model to capture features at varying resolutions, facilitating accurate segmentation of both fine-grained details and broader anatomical structures. Removing this block caused the Dice Index to drop to $89.80 \pm 0.25\%$ and the Boundary F1 Score to $86.90 \pm 0.30\%$, as shown in Table 19. Classification metrics, as detailed in Table 20, reflected a reduction in Recall from $94.30 \pm 0.10\%$ to $92.40 \pm 0.12\%$.

The absence of multi-scale processing significantly impacted the model's ability to handle diverse anatomical structures. For instance, detailed segmentation of fetal limbs requires high-resolution features, whereas larger structures like the head circumference require broader contextual understanding. This limitation can lead to inaccuracies in measurements such as HC and AC, potentially misguiding clinical decisions related to fetal weight estimation or growth monitoring.

4.10.4. Impact of Gradient-Based adaptive weighting optimization Block

The Gradient-Based Adaptive Weighting Optimization Block

Table 19
Impact of Multi-Scale Feature Extraction on Segmentation Metrics.

Model	Dice Index (%)	HD (mm)	Boundary F1 (%)	MAE (mm)	PMAE (%)
DTNAU-Net w/o Multi-Scale	89.80 ± 0.25	6.80 ± 0.30	86.90 ± 0.30	1.18 ± 0.03	5.20 ± 0.12
Proposed DTNAU-Net	91.12 ± 0.20	5.03 ± 0.15	88.50 ± 0.20	0.90 ± 0.01	4.40 ± 0.10

Table 20
Impact of Multi-Scale Feature Extraction on Classification Metrics.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DTNAU-Net w/o Multi-Scale	92.80 ± 0.20	92.60 ± 0.15	92.40 ± 0.12	92.50 ± 0.14
Proposed DTNAU-Net	94.85 ± 0.15	94.50 ± 0.12	94.30 ± 0.10	94.30 ± 0.12

dynamically balances learning across segmentation and classification tasks by analyzing gradient magnitudes and adjusting task-specific loss weights in real-time. This mechanism helps maintain stable optimization when multiple tasks compete during training. The impact of removing this block on segmentation performance is summarized in Table 21. Without the gradient-based weighting mechanism, the segmentation Dice Index decreased to $89.90 \pm 0.22\%$ and the Boundary F1 Score dropped to $87.00 \pm 0.25\%$, indicating reduced boundary delineation accuracy.

As shown in Table 21, incorporating the Gradient-Based Adaptive Weighting Optimization Block significantly improves segmentation accuracy, boundary localization, and biometric measurement precision. Similarly, the impact on classification performance is presented in Table 22. Removing the block reduced classification Accuracy to $93.10 \pm 0.18\%$ and F1 Score to $92.90 \pm 0.15\%$, indicating reduced reliability in plane recognition.

As indicated in Table 22, the proposed model achieves higher classification performance when the adaptive weighting mechanism is included.

Without the Gradient-Based Adaptive Weighting Optimization Block, the model showed imbalanced learning, particularly in challenging scenarios requiring simultaneous segmentation and classification. For instance, identifying fetal ultrasound planes while delineating their boundaries became less effective, reducing the reliability of combined outputs. The dynamic balancing enabled by this block is critical for ensuring both segmentation accuracy and classification precision in comprehensive fetal assessments, where simultaneous optimization of multiple tasks is essential.

4.10.5. Impact of dual-task learning

To directly evaluate the contribution of simultaneous classification and segmentation, an additional ablation experiment was conducted by removing the classification branch and training the model exclusively for segmentation. All architectural components (Residual Blocks, Dual Attention, Multi-Scale Extraction, and Gradient-Based Adaptive Weighting) were retained to ensure fair comparison. The resulting segmentation performance is summarized in Table 23.

As shown in Table 23, the inclusion of the classification branch improves segmentation performance across all metrics, including Dice Index, Boundary F1 Score, and MAE. Performance differences were statistically validated using a paired *t*-test at a 95% confidence level ($p < 0.01$), confirming that the observed improvement is statistically significant. These results highlight that dual-task learning provides complementary information between segmentation and classification,

Table 21
Impact of Gradient-Based Adaptive Weighting Optimization Block on Segmentation Metrics.

Model	Dice Index (%)	HD (mm)	Boundary F1 (%)	MAE (mm)	PMAE (%)
DTNAU-Net w/o gradient-based block	89.90 ± 0.22	6.20 ± 0.18	87.00 ± 0.25	1.17 ± 0.03	5.20 ± 0.10
Proposed DTNAU-Net	91.12 ± 0.20	5.03 ± 0.15	88.50 ± 0.20	0.90 ± 0.01	4.40 ± 0.10

Table 22
Impact of Gradient-Based Adaptive Weighting Optimization Block on Classification Metrics.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DTNAU-Net w/o gradient-based block	93.10 ± 0.18	92.80 ± 0.15	93.00 ± 0.14	92.90 ± 0.15
Proposed DTNAU-Net	94.85 ± 0.15	94.50 ± 0.12	94.30 ± 0.10	94.30 ± 0.12

Table 23
Impact of Dual-Task Learning on Segmentation Performance.

Model Configuration	Dice Index (%)	HD (mm)	Boundary F1 (%)	MAE (mm)	PMAE (%)
Segmentation-Only (No Classification Branch)	90.40 ± 0.22	5.60 ± 0.18	87.10 ± 0.24	1.05 ± 0.02	4.85 ± 0.11
Proposed DTNAU-Net (Dual-Task)	91.12 ± 0.20	5.03 ± 0.15	88.50 ± 0.20	0.90 ± 0.01	4.40 ± 0.10

allowing the network to learn more discriminative and robust feature representations. In the context of fetal ultrasound, this synergy enhances boundary delineation while simultaneously reinforcing plane recognition, ultimately improving overall model reliability for clinical biometric assessments.

4.10.6. Combined Impact of all components

To understand the combined effect of architectural components, the full DTNAU-Net model was compared against its variants where one or more components were removed. The results, summarized in Tables 24 and 25, demonstrate that the full model consistently achieved superior segmentation and classification performance.

- **Segmentation Results:** The Dice Index for the full model was $91.12 \pm 0.20\%$, significantly outperforming its closest variant. The Boundary F1 Score and HD metrics further confirmed the full model's ability to delineate anatomical boundaries with precision.
- **Classification Results:** The Accuracy of the full model reached $94.85 \pm 0.15\%$, reflecting its balanced performance across both tasks. Precision, Recall, and F1 Score were also the highest among all variants, underscoring the combined effectiveness of its components.

These findings validate the design of DTNAU-Net, emphasizing the synergy between Residual Blocks, Dual Attention Mechanisms, Multi-Scale Feature Extraction and Gradient-Based Adaptive Weighting Optimization. Together, these components ensure robust performance in

Table 24
Combined impact of all components on segmentation metrics.

Model	Dice Index (%)	HD (mm)	Boundary F1 (%)	MAE (mm)	PMAE (%)
DTNAU-Net w/o Residual Blocks	89.02 ± 0.25	6.50 ± 0.22	86.70 ± 0.30	1.20 ± 0.03	5.30 ± 0.10
DTNAU-Net w/o Dual Attention	90.10 ± 0.22	6.20 ± 0.20	87.10 ± 0.25	1.15 ± 0.02	5.10 ± 0.10
DTNAU-Net w/o multi-scale	89.80 ± 0.25	6.40 ± 0.21	86.90 ± 0.30	1.18 ± 0.03	5.20 ± 0.10
DTNAU-Net w/o gradient-based block	89.90 ± 0.22	6.20 ± 0.18	87.00 ± 0.25	1.17 ± 0.03	5.20 ± 0.10
DTNAU-Net w/o Attention	90.50 ± 0.22	5.70 ± 0.18	87.20 ± 0.25	1.10 ± 0.02	4.70 ± 0.10
Proposed DTNAU-Net	91.12 ± 0.20	5.03 ± 0.15	88.50 ± 0.20	0.90 ± 0.01	4.40 ± 0.10

Table 25

Combined impact of all components on classification metrics.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DTNAU-Net w/o Residual Blocks	92.50 ± 0.20	92.10 ± 0.18	92.20 ± 0.16	92.15 ± 0.17
DTNAU-Net w/o Dual Attention	93.20 ± 0.18	92.80 ± 0.15	92.70 ± 0.14	92.75 ± 0.14
DTNAU-Net w/o multi-scale	93.00 ± 0.18	92.70 ± 0.16	92.60 ± 0.15	92.65 ± 0.15
DTNAU-Net w/o gradient-based block	93.10 ± 0.18	92.80 ± 0.15	93.00 ± 0.14	92.90 ± 0.15
DTNAU-Net w/o Attention	93.80 ± 0.15	93.40 ± 0.12	93.30 ± 0.12	93.35 ± 0.12
Proposed DTNAU-Net	94.85 ± 0.15	94.50 ± 0.12	94.30 ± 0.10	94.30 ± 0.12

segmentation and classification tasks, making DTNAU-Net a reliable tool for fetal biometric analysis in clinical applications.

4.11. Statistical analysis

A comprehensive statistical analysis was conducted to evaluate DTNAU-Net's effectiveness compared to state-of-the-art models, including U-Net (Ronneberger et al., 2015), U-Net++ (Zhou et al., 2018), ResUNet++ (Jha et al., 2019), Deeplabv3+ (Chen et al., 2018), Double U-Net (Jha et al., 2020), MSRF-Net (Srivastava et al., 2022), FetSAM (Alzubaidi et al., 2024), FetalNet (Plotka et al., 2021), TransUNet (Chen et al., 2021), and nnUNet (Isensee et al., 2021). The assessment employed the Paired T-test, Wilcoxon Signed-Rank Test, and Friedman Test, analyzing segmentation and classification metrics (Dice Index, Hausdorff Distance, Accuracy, Precision, Recall, and F1-Score) across the HC18 and Zenodo datasets.

4.11.1. Paired t-test

A paired *t*-test (Warfield et al., 2004) ($\alpha = 0.05$) was conducted using fold-wise performance values obtained from five-fold patient-level cross-validation. On the HC18 dataset, DTNAU-Net achieved a Dice Index of $93.4\% \pm 0.85$ and a Hausdorff Distance of 3.12 ± 0.20 mm, while on the Zenodo dataset it achieved a Dice Index of $92.8\% \pm 0.88$ and a Hausdorff Distance of 3.18 ± 0.21 mm. These dataset-specific results were statistically superior to TransUNet ($p = 0.022$) and nnUNet ($p = 0.020$).

For classification, DTNAU-Net achieved an accuracy of $90.5\% \pm 0.8$ (HC18) and $90.0\% \pm 0.8$ (Zenodo), with corresponding F1-scores of $90.5\% \pm 0.8$ and $90.0\% \pm 0.8$. All improvements were statistically significant ($p < 0.05$). These values represent dataset-specific fold-wise results, whereas the overall cross-dataset performance summarized in Table 3 corresponds to the aggregated mean across datasets.

Detailed statistical comparisons for all models and metrics are provided in Supplementary Table S1.

4.11.2. Wilcoxon signed-rank test

The Wilcoxon Signed-Rank Test (Rosner et al., 2006) (non-parametric, $\alpha = 0.05$) confirmed statistically significant improvements ($p < 0.05$) for DTNAU-Net. On HC18, DTNAU-Net achieved a Dice Index of $93.4\% \pm 0.85$ and HD of 3.12 ± 0.20 mm, with rank sums (160 vs. 1 negative) indicating consistent advantages. Classification accuracy was $90.5\% \pm 0.8$ ($p < 0.01$).

These results align with the paired *t*-test findings, reinforcing DTNAU-Net's robustness across imaging conditions. Further statistical details are in Table S2.

4.11.3. Friedman test

A Friedman Test (Liu & Xu, 2022) (non-parametric, $\alpha = 0.05$) ranked DTNAU-Net as the top-performing model across all segmentation and classification metrics. Post-hoc analysis confirmed significant

differences ($p < 0.01$) between DTNAU-Net and the second-ranked models. These findings reinforce DTNAU-Net's reliability in fetal ultrasound analysis (refer Supplementary Table S3).

4.12. Explainable AI analysis

To enhance interpretability, Explainable AI (XAI) techniques were applied to DTNAU-Net's predictions for segmentation and classification. Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) provided insights into feature importance and decision-making processes.

4.12.1. Shapley Additive Explanations (SHAP)

SHAP (Salih et al., 2025) analysis identified pixel intensity and edge features as primary contributors to segmentation, particularly for fetal head and femur contours. Classification relied on texture patterns and shape descriptors. Summary plots in Supplementary Figs. S5 and S6 show DTNAU-Net prioritizing anatomically relevant regions while suppressing noise. Key SHAP values include: Pixel intensity: 0.75 (Segmentation), Edge features: 0.65 (Segmentation) and Texture features: 0.55 (Classification). These findings validate the model's reliability for biometric measurements (refer Supplementary Table S4).

4.12.2. Local Interpretable Model-Agnostic Explanations (LIME)

LIME (Salih et al., 2025) provided local interpretability, confirming DTNAU-Net's focus on key anatomical structures. Key impact scores include Edge features: 0.72 (segmentation) and Texture relevance: 0.66 (classification). Visualizations and impact scores (Figs. S7 and S8, Table S5) reinforce DTNAU-Net's transparency and reliability for fetal growth monitoring. LIME and SHAP collectively enhance trust in its decision-making process for clinical applications.

5. Conclusions

This study introduced DTNAU-Net, an advanced deep learning model designed for precise and efficient fetal biometric segmentation and classification of fetal ultrasound planes. Through rigorous experimentation, we established that the integration of Residual Blocks, Dual Attention Mechanisms, Multi-Scale Feature Extraction, and Gradient-Based Adaptive Weighting Optimization significantly improved segmentation accuracy. The DTNAU-Net significantly reduced inference time, enabling near real-time ultrasound analysis with total processing time of ~ 0.12 s for fetal biometric assessments. Furthermore, explainable AI methods (SHAP, LIME) reinforced the reliability and transparency of its decision-making process, making it a trustworthy tool for clinical adoption. These improvements may be useful to enhance clinical workflow efficiency, minimize patient wait times, and facilitate AI-assisted diagnostics in resource-limited settings. Future work will focus on optimizing model generalizability and integrating real-time deployment into ultrasound systems to advance prenatal diagnostics.

From a clinical deployment perspective, the decoupled design of DTNAU-Net avoids the failure mode of incorrect classification leading to incorrect segmentation or biometric estimation. Classification confidence can instead support frame validation or quality control, while segmentation remains stable and independent. Although cross-dataset validation demonstrates promising generalization performance, future multi-center prospective validation across diverse imaging devices and operators will further strengthen clinical deployment readiness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

All authors extend the acknowledgment for the logistical support provided by MNNIT Allahabad, Prayagraj during the completion of the study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eswa.2026.132017>.

Data availability

Data will be made available on request.

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